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NATIONAL ENVIRONMENTAL ACTION PLAN

1992-1996

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Ministry of Environment and Parliamentary Affairs
October 1991

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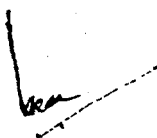
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**MESSAGE FROM
HONOURABLE MINISTER OF ENVIRONMENT
AND PARLIAMENTARY AFFAIRS**

We have had many ambitious economic development programmes in the forty three years, since Independence. Until recently, there has been inadequate attention paid to the environmental aspects. However today, we are firmly committed to the concept of sustainable development and the need to protect and manage the environment. We are fully alert to the consequences of ignoring the environmental impacts of social and economic activities.

The enactment by Parliament of the National Environmental Act of 1980, the establishment of the Central Environmental Authority in 1981 under that Act, and the Environmental Council in 1982, the establishment of District Environmental Agencies in 1984, the Amendment to the National Environmental Act in 1988, the adoption of the National Environmental (Protection and Quality) Regulations of 1990, and above all, the creation in 1990 of a Ministry of Cabinet rank to be in-charge of the subject of environment, are important National landmarks in the progress towards legislative and institutional development in the field of environment.

The National Environmental Action Plan marks another major advance. This five year plan is based on national priorities and on an assessment of resource availability. It is anticipated that the components of the plan will soon be incorporated into the annual work plans of the agencies responsible for the various sectors covered. I have no doubt that the National Environmental Action Plan will set the environmental agenda for the next five years and help to ensure that we are firmly on the path of environmentally sustainable development.



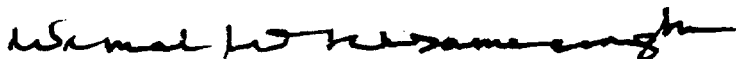
**(M VINCENT PERERA)
MINISTER OF ENVIRONMENT AND
PARLIAMENTARY AFFAIRS**

**MESSAGE FROM
HONOURABLE MINISTER OF ENVIRONMENT**

It is with great pleasure and sincere dedication that we, the Government of Sri Lanka (GOSL), together with the Ministry of Environment, the Central Environmental Authority (CEA) and all other allied agencies, present this National Environmental Action Plan (NEAP), a document that has been substantially prepared by Sri Lankan resource persons identifying local environmental issues and making efforts for providing solutions to such issues. This Action Plan can well be considered as a graduation of knowledge acquired through extensive studies in the field of environment. In other words, this NEAP contains the core contents of: (a) the Action Plan of the National Conservation Strategy (NCS) prepared by the CEA under the patronage of the Ministry of Environment and with an effective feedback from all who matter in the field of environment, including NGOs; (b) the Environmental Action Plan prepared by the GOSL together with the World Bank; and (c) the Sri Lanka National Report to the United Nations Conference on Environment and Development (UNCED) in 1992 in Brazil.

It should be mentioned that the origin of the NEAP stems from the efforts of the World Commission on Environment and Development which put out its path-breaking Report, *OUR COMMON FUTURE* (1987). Consequent upon the publication of this Report, the International Union for Conservation of Nature and Natural Resources (IUCN) went ahead with the responsibility of alerting Governments to the urgent need for environmental conservation and development, particularly through its document titled *The World Conservation Strategy*. The GOSL has been among the first to recognize the objectives laid down by the IUCN, and on a proposal made by His Excellency Ranasinghe Premadasa as the then Prime Minister, the former President of Sri Lanka appointed, in 1982, a Task Force for the preparation of a NCS for Sri Lanka.

In consequence, Sri Lanka's NCS which came out as a blue print in December 1988 was accepted by the Government as the declaration of National Policy in relation to the conservation of nature and natural resources in the country. The next step needed in the process was the transformation of this Strategy from a general statement of programmes and principles to a set of policy guidelines and action-oriented plan. Through a mechanism of steering committees and working groups, a draft Action Plan of the NCS was developed by the CEA, after discussion with the Ministry of Environment, the Environmental Council, the NGOs and the Secretaries of the allied Ministries. In the meantime, with the assistance of the Government of Sri Lanka, the staff of the World Bank initiated preparing an Environmental Action Plan, known as EAP, embodying major recommendations of the draft Action Plan of the NCS and concentrating on a few major environmental issues. Then a policy decision was taken for the preparation of one single Action Plan embodying not only the core contents and recommendations of the Action Plan of the NCS and EAP but also those of the country's National Report prepared for the UNCED to be held in Brazil in July 1992. The final outcome of all these efforts is the present NEAP for which a foreward is written by me as Minister of Environment. I have no doubt that this Action Plan will serve as an important guidebook for introducing environmental aspects to the country's development process which has now successfully entered into a take-off stage.



(DR WIMAL WICKRAMASINGHE)

MINISTER OF ENVIRONMENT AND

MINISTER OF STATE FOR POLICY PLANNING AND IMPLEMENTATION

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PREAMBLE

Sri Lanka is endowed with a rich natural environmental heritage in which, from time immemorial, our civilization has flourished and our cultural values have developed. The rapid growth in population over the last few decades and their rising demands are fast depleting our natural resource base and limiting the development prospects of the island. There have been many development plans which neglected or served to undermine the environment. In turn there have been many programmes and strategies for the protection and conservation of our environment. There was very little done to inject environmental concerns into economic development programmes or to incorporate economic development objectives into environment conservation programmes. However, there is now growing concern that economic progress and environmental protection should be complementing objectives and that these should be developed in a balanced and integrated manner, with in the scope of formulating and implementing of national and sub-national plans. Among the important components are rapid industrialization, enhancing agriculture productivity and expanding infrastructural facilities. These programmes should be harmonious with the environment, if the development process is to be sustained to benefit future generations.

This National Environmental Action Plan (NEAP) - 1992-1996 is the first comprehensive time phased environmental planning document of Sri Lanka. Its objective is to establish an environmental agenda within the development context. The plan draws extensively from recent publication, namely the *National Report* for the United Nations Conference on Environment and Development funded by UNDP and co-ordinated by M/E&PA, the *National Conservation Strategy - Draft Action Plan*, formulated by the Central Environmental Authority, the *Environmental Action Plan* funded by the World Bank and co-ordinated by M/E&PA, the *Natural Resources of Sri Lanka - Conditions and Trends* funded by USAID and published by the Natural Resources, Energy and Science Authority, and reports and sectoral papers prepared by various Government agencies in co-operation with key donors. The NEAP has been compiled by a group of eminent specialists, many of whom were extensively involved in the preparation of the above mentioned publications. Guidance and co-ordination during the preparatory stages was provided by the National Environmental Steering Committee (NESC), which is chaired by the Secretary to the Ministry of Policy Planning and Implementation and co-chaired by the Secretary to the Ministry of Environment and Parliamentary Affairs. This committee includes the Secretaries of all relevant Ministries. A unique feature of the NEAP is the level of involvement of the non-governmental sector in development of the plan. The suggestions made by the state agencies, non-governmental organisations and the public at fora organized by M/E&PA provided valuable inputs both into the evolution of the basic structure of the plan and into the detailed component of the various sections of the structure.

The general format of the NEAP is somewhat different from that of most planning documents in Sri Lanka. Specific actions to be undertaken in sectors are given briefly with an indication of the required time frame and the level of investment needed. These estimates at best are only tentative, subject to variation depending on the changes in the socio-economic context.

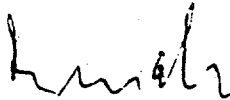
The NEAP includes both preventive and corrective measures in areas such as water and land resources, bio-diversity and wildlife, forestry, urban pollution, industrial pollution, coastal resources, energy, education and culture. The Institutional enhancement required to implement the different actions have also been comprehensively reviewed. The financing of the different actions is to be met from both internal and external funds. The implementation stage of NEAP will involve a multitude of agencies, and co-ordination will be effected through the NESC.

The NEAP envisages an investment requirement of around four billion rupees (or one hundred million dollars). This is ambitious but not expensive in comparison with the investments made on development plans during the last decade. The Mahaweli Project alone costs over fifty billion rupees. Overall the NEAP would account for less than 5% of the total development budget. The investments in the environmental field are intended to improve the quality of the overall development process and its long term sustainability. Therefore, environmental planning should not be considered as an extra burden but as a vital cost-effective and integral component of the national development strategy.

The actions recommended for each area indicate a certain degree prioritization in terms of timing. The process of prioritization was generally guided by the following criteria:

- 1 The degree of public concern as reflected in public discussions;
- 2 The need to preserve certain non-renewable resources and to prevent irreversible damage;
- 3 Likely economic and social impact;
- 4 The degree of coverage by existing programmes; and
- 5 Practical feasibility in terms of resources, personal and other constraints.

Environmental Planning is a continuous activity in which lessons learnt should be fed back into the planning process. Therefore, the Ministry will continuously monitor and evaluate the implementation of this plan. Moreover, five years is much too small a time-frame for the achievement of many environmental goals. But the base documents from which the NEAP has evolved, particularly the NCS Action Plan, contain material for a wealth of proposals which could not be squeezed into the five year framework of this 1992-1996 Plan. Work on the second five year Environmental Plan (1997-2001) must commence in the early part of the nineties. This task will be high on the priority list of the Ministry of Environment and Parliamentary Affairs in the years to come.



DEVANESAN NESIAH
Secretary
Ministry of Environment
and Parliamentary Affairs

October 4, 1991.

ABBREVIATIONS AND ACRONYMS

AD	-	Agriculture, Department of
ADB	-	Asian Development Bank
AGA	-	Assistant Government Agent
CCD	-	Coast Conservation Department
CEA	-	Central Environmental Authority
CEB	-	Ceylon Electricity Board
CISIR	-	Ceylon Institute of Scientific and Industrial Research
DWLC	-	Department of Wildlife Conservation
EAP	-	Environmental Action Plan
EIA	-	Environmental Impact Assessment
FAO	-	Food and Agricultural Organization
FD	-	Forest Department
GCEC	-	Greater Colombo Economic Commission
GOSL	-	Government of Sri Lanka
ID	-	Irrigation Department
IDA	-	International Development Association (World Bank affiliate)
IEE	-	Initial Environmental Examination
IUCN	-	International Union for the Conservation of Nature
JEDB	-	Janatha Estates Development Board
LUPPD	-	Land Use Policy and Planning Division
MEIP	-	Metropolitan Environment Improvement Programme
M/AD&R	-	Ministry of Agricultural Development and Research
M/CA	-	Ministry of Cultural Affairs
M/E&PA	-	Ministry of Environment and Parliamentary Affairs
M/IS&T	-	Ministry of Industry, Science and Technology
M/LI&MD	-	Ministry of Lands, Irrigation and Mahaweli Development
M/SS	-	Ministry of Social Services
MOE	-	Ministry of Environment (Project Ministry)
M/PP&I	-	Ministry of Policy Planning and Implementation
MT	-	Metric Ton
NARA	-	National Aquatic Resources Agency
NAREPP	-	Natural Resources and Environmental Policy Project
NARESA	-	National Resources, Energy and Science Authority
NBRO	-	National Building Research Organization
NCS	-	National Conservation Strategy
NESC	-	National Environment Steering Committee
NFCP	-	National Fuelwood Conservation Program
NGO	-	Non-Governmental Organization
NLC	-	National Land Commission
NWSC	-	National Wetland Steering Committee
PAA	-	Project Approving Agencies
PIP	-	Public Investment Program
PME	-	Public Manufacturing Enterprises

SAM	-	Special Area Management
SGC	-	State Gem Corporation
SLIDA	-	Sri Lanka Institute for Development Administration
SLSPC	-	Sri Lanka State Plantations Corporation
STC	-	State Timber Corporation
TSHDA	-	Tea Small Holdings Development Authority
UDA	-	Urban Development Authority
UNCED	-	United Nations Conference for Environment and Development
UNDP	-	United Nations Development Programme
USAID	-	United States Agency for International Development

CURENCY EQUIVALENTS

(In Sri Lanka Rupees - As at September 30, 1991)

Sterling Pounds	=	70.50
US Dollars	=	41.84
Japanese Yen	=	00.30
Deutch Mark	=	23.94
French Frank	=	07.02

LAND RESOURCES

OVERVIEW

Sri Lanka is predominantly rural and agrarian. Nearly 80 per cent of the people live in the country side and over 45 per cent depend on agriculture for a living. Within its total land area of 6.56 million hectares, a population of a little over 17 million live, making it one of the most densely peopled countries even within Asia. With increasing population, land:man ratios have declined from about 2.7 ha/head to 0.38 ha/head within the last hundred years. Due to the slow growth of the economy and the lack of avenues of employment outside agriculture, the proportion of the population engaged in agriculture decreased only slightly, by about 5 per cent during the last four decades.

The extent of land under agriculture remained more or less static with an increase of only 2 per cent since National Independence. The land under natural forest cover which was around 70 per cent of the total land area in 1900 was reduced to 44 per cent by 1956 and is about 22 per cent (or 1.5 million ha.) today. Over-exploitation of some of the existing agricultural lands and encroachment of some of the remaining state land under forest has led to an accelerated rate of land degradation creating a variety of adverse ecological consequences such as soil erosion, siltation, landslides and loss of bio-diversity.

Some of the main environmental issues related to land are as follows:

- (1) Loss of productivity of agricultural lands through soil erosion;
- (2) Continuing practice of 'chena' cultivation which despite restrictions still covers nearly 15 per cent of the land area;
- (3) Encroachment of hydrologically critical areas such as streams and reservoir reservations;
- (4) Degradation of the productivity and profitability of large extents of tea lands in the plantation sector;
- (5) Insecurity of tenure and lack of incentives to develop and manage land by small holders;
- (6) Inadequate landuse planning and implementation at the national, provincial and local levels;
- (7) Poor enforcement of the relevant legislation due to lack of political will and fragmentation of responsibilities among a multitude of state agencies;
- (8) Increasing damage to life and property due to natural disasters such as landslides and floods;
- (9) Inadequate attention paid to the conservation of land at high elevations (above 1500 metres) resulting in adverse consequences to the whole country.

1

LOSS OF PRODUCTIVITY OF AGRICULTURAL LANDS THROUGH SOIL EROSION**BACKGROUND:**

The rates of soil erosion in most agricultural areas and certain development sites of Sri Lanka are high by any standard. The problem of soil loss exists in all agro-ecological zones, although its intensity can vary according to the climate, slope and the ground cover. The worst affected area is the mid-country which falls within an elevational range of 300-1000m. Estimates of soil loss from most affected areas such as those under tobacco grown on steep slopes are reported as high as 388-913 ha/year (TAMS, 1980). The rapid rate of soil loss is reflected in the high silt loads carried by rivers which eventually deposit them in reservoir beds. Some studies indicate that on the average 0.5 million tons of sediment passed through Mahaweli Ganga at Peradeniya gauging station annually. About 70 per cent of this load was trapped at Polgolla reservoir which was silted up to 45 per cent of its capacity within the first 12 years of its commissioning.

The implementation machinery of the Soil Conservation Act of 1951 which performed a useful role in identifying and declaring erodible areas and taking corrective measures has virtually ceased to function. The Soil Conservation Division of the Department of Agriculture no longer exists. The devolution of functions of the Department to Provincial Councils without adequate preparation at the provincial level had caused further deterioration of this situation. There are some current efforts to introduce new legislation, although there is no single agency to take over the implementation tasks. The incentives provided to small holders to invest on soil conservation by state agencies such as the Tea Small Holding Development Authority (TSHDA) and the tobacco companies are extremely inadequate. In these circumstances, soil erosion at least in the critical watersheds of some mid-country reservoirs cannot be mitigated without some concerted effort and massive investments. Although the damage caused by soil erosion is often obvious in the loss of land productivity and in high siltation rates, there is a need to understand more fully the economics of soil conservation.

The main issues related to soil conservation are:

- i Deficiencies in the legislation and lack of clear institutional responsibility;
- ii Lack of adequate incentives to small farmers for soil conservation;
- iii Low priority accorded to soil conservation, particularly by the Ministries of Agriculture, at the centre as well as at the Provinces; and
- iv High market prices fetched by erosion enhancing crops such as tobacco, vegetables and potatoes compared with those fetched by tree crops.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Draft a new Soil Conservation Act which would supersede that of 1951 and implement its provisions with proper monitoring and evaluation.	Immediate	3,000,000	M/E&PA, CEA
2 Increase incentives to small farmers for soil conservation and implement this scheme through non-governmental farmer organizations.	Medium	250,000	TSHDA, Provincial Ministries of Agriculture
3 Provide assistance to PCs in 'Erodible Areas' for soil conservation purposes.	Long-term		PCs of Central Province, Uva and Sabaragamuwa
4 Re-locate and provide alternative lands and employment opportunities for farmers cultivating seasonal crops on encroached erosion prone lands.	Immediate	1,000,000	Provincial Ministries and M/LI&MD
5 Ensure that tobacco companies and other private enterprises operating in erosion prone areas re-invest an adequate proportion of their profits on soil conservation.	Immediate	100,000	M/E&PA, M/LI&MD

CONTINUING PRACTICE OF CHENA CULTIVATION

BACKGROUND:

Chena or shifting cultivation covers about 1.2 million hectares or about 15 per cent of the total land area or nearly half the total extent under agricultural lands. The extent of chena lands has apparently increased by about 120,000 ha. between mid-1950s and the early 1980s according to available estimates. This was despite governmental restrictions and a declared embargo on shifting cultivation in recent years. The chena is practised largely on available State lands under forests either by individuals or occasionally by more organised persons with business interests.

The detrimental ecological effects of chena cultivation were recognized for a long time. It depletes soil fertility, causes weed infestation, soil erosion and crop yield decline. Successive governments since the colonial period have attempted to control it administratively and legislatively with little success. Chenas produce over 80 per cent of Sri Lanka's rainfed grains and around a quarter million families depend on them for livelihood. Most chena farmers come from the poorest segments of society, and many of them have no easy access to any other means of livelihood. The popularly elected Governments therefore, find it difficult to deal with chena cultivators too harshly. The chena represents an interesting example of that problematic link between poverty and environmental conservation. It is now clear that legislative or administrative action to control chena farming can have only limited success without the development of alternative strategies of gainful employment. Chena work is strenuous and provide no security of tenure and many farmers will be glad to leave it for other avenues of employment and income are provided.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 As an interim measure, permit chena cultivation to continue on certain identified lands and subject to the readoption of certain traditional chena practices which are less harmful to the environment than current practices, and the adoption of certain modern methods of Conservation Farming.	Immediate	100,000	AD (Maha Iluppallama) Mass media
2 Develop a forest rehabilitation strategy for those degraded forests due to chena farming with community participation.	Long-term	1,000,000	NGOO, Farmer Orgns., FD
3 Map out selected areas of existing degraded forest lands with high agricultural potential and convert them for agricultural use, so that a considerable proportion of chena farmers can be absorbed into permanent cultivation on these lands. The lands available in the Dry and Intermediate Zones for agriculture purposes is estimated to be over 600,000 ha. This is a major task that would virtually involve the redrawing of the forest map in these areas.	Long-term	2,500,000	M/LI&MD, LUPPD, FD

ENCROACHMENT OF HYDROLOGICALLY CRITICAL AREAS**BACKGROUND:**

The law (Crown Lands Ordinance and its Regulations) provides for the reservation of belts of land on stream banks with widths varying according to the size of streams. The protection of these reservations is one of the many functions of the Government Agents. However, over the years many of these reservations have been encroached on by unauthorized persons for agriculture and even for permanent settlement. In one of the early field surveys by the Water Resources Board (1969) it was found that only about 17 per cent of such reservations have been spared in one administrative division of the Badulla District. Evidently the existing laws and institutions have failed to give adequate protection to hydrological reservations.

The neglect of these reservations has resulted in siltation, bank erosion and even landslides. Encroachments may also affect the flow of springs and streams and their function as buffer zones between sloping lands and the water courses, which not only absorb the impacts of erosion, but even agro-chemicals used on high lands. The social and economic costs of these impacts are hard to quantify, but it may be worth attempting to do so.

Similarly, irrigation canal and tank reservations have also been heavily encroached. The irrigation laws provide for their maintenance, but their implementation has been very weak. The current irrigation regulations require a belt of land around the inundation area of each reservoir upto the bund-top-level (BTL) contour to be protected. It is well known that settlements and even permanent structures have come up within these critical zones. There is hardly any effective plan of action to protect even the newly created Mahaweli reservoirs. This has resulted in high rates of siltation and the diminution of water supplies for irrigation and hydropower generation.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Up to 10 per cent of the land area of the country may come under hydrological reservations in terms of existing laws. Their protection is however, not practical in many areas where settlements and townships have been developed on river banks. A more pragmatic approach may include following actions: i Preparation of an inventory of all reservations prescribed by law, separately identifying those which have not been encroached on; ii The undertaking of a review of the prevailing legal and institutional provisions for the protection of reservations; iii The development and implementation of a programme of action to restore reservations encroached on, where feasible; iv Creation of "green belts" on reservoir reservations.	Immediate Immediate Medium Long-term	100,000 50,000 100,000 1,000,000	WRB, Universities CEA, EFL GAs and Regional Directors of Irrigation ID, NGOs

4

DEGRADED TEA LANDS**BACKGROUND:**

At present there is about a quarter million hectares of land under tea. The large proportion of these lands are under old seedling tea with low yields, often located in erosion prone lands. The Tea Commission of 1968 excluded some 160,000 ha. of such tea lands from their programme of planting with high yielding VP tea. The Commission predicted that, a good proportion of this tea land will become increasingly uneconomical. After the Land Reforms in the 1970s, such lands had been largely neglected or left uncultivated, particularly by private owners. Thus there was a loss of over 10 per cent of the total tea acreage between 1962 and 1982 particularly in the mid-country. At present, up to 30 per cent of the tea land can be considered as marginal or uneconomical.

When there is a well-maintained good tea cover it can prevent soil erosion as by a dense natural forest. When it is degraded it can record some of the highest soil erosion rates for any crop. Some of the uneconomical tea lands can therefore, be rehabilitated, but the majority may have to be converted to other land uses. Uneconomical tea land is also a burden on the tea industry, since these have to be maintained with earnings from more productive lands.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Mapping of marginal tea lands, identifying and classifying lands that can be and cannot be rehabilitated.	Immediate	100,000	LUPPD, MASL (Env't Unit)
2 Developing a programme of action that would encourage the return of uneconomical tea lands to natural forests, plantation forestry, or small holder land use with tree crops, etc.	Long-term	500,000	NADSA, NGOO
3 Developing a strategy to support the two major Plantation Corporations, i.e., the JEDB and the SLSPC to undertake crop diversification and soil conservation within their lands.	Long-term	100,000	JEDB, SLSPC (Soil Conservation Committee)
4 Providing adequate incentives for the small holders to invest more on soil conservation.	Long-term	1,000,000	TSHDA, AD

5 | LAND TENURE AND LAND DEGRADATION

BACKGROUND:

Over 80 per cent of the total land area of Sri Lanka still remains under the ultimate ownership of the government, despite the massive land alienation programmes which transferred over a million hectares by 1990 to the peasantry and others in need of land. This is in clear contrast to the land tenure situation in many countries. The land reforms of the 1980s brought more land under state ownership. However, the bulk of state lands are in forest or wildlife reserves and in the Dry and Intermediate Climatic Zones. The State monopoly of land and the resultant imbalance between the demand for and supply of land eventually led to wide spread encroachments on State lands. It was estimated that by the time of the encroachment survey of 1979, about half a million persons have encroached upon 376,938 ha. of land covering some 6 per cent of the total land area of the country.

The Government adopted a policy of encroachment regularization over the years, and today such regularization account for over a quarter of all lands alienated. Although encroachments on reservations were not regularized, the great majority of such encroachers continue to occupy these lands on stream, canal and reservoir reservations. The lenient attitude towards encroachers by the successive Governments after independence has undoubtedly induced more encroachments and encouraged many villagers to think that the easiest way to obtain a block of State land is by encroachment.

A considerable proportion of encroached lands are in ecologically fragile environments such as stream banks, steep slopes and reservoir margins. Due to tenurial insecurity, many encroachers do not take an interest in protecting their lands from erosion and other forms of degradation. their main objective is to exploit the land as much as possible before they are ejected.

There have been suggestions that encroachers and chena farmers should be treated as innovative "pioneers" and be given credit facilities although this would encourage further encroachments. A more feasible suggestion is to convert all regularized lands which are now under LDO permits to free titles making them readily marketable properties. This had been discouraged in the past due to fears of 'improvident alienation' by poor allottees who may dispose of their lands under distress and return to the status of landless peasantry or migrate to cities in search of employment. These are hard questions to resolve but vital policy decisions need to be taken at least on an experimental basis.

Traditionally, encroachments were regularized through the provision of a permit under the Land Development Ordinance (LDO). Until the introduction of the Swarnabhoomi Programme the ultimate ownership of the LDO lands on which nearly a quarter of the population live, rested with the Government. Even the Swarnabhoomi grant is conditional since it has some restrictions on sub-division and transfer. Since the LDO and Swarnabhoomi lands are not readily marketable, the commercial banks are reluctant to consider such lands for security in providing credit for long term land development investments such as soil erosion. Since credit is more easily available on a short term basis for cultivation crops which induce erosion than on a long-term basis for tree crops or land development, the net effect on soil conservation is negative.

In certain parts of the wet zone, traditional tenurial patterns such as the tattumaru system and ande tenancy tends to create insecurity of tenure through clouded titles and encumbered ownership rights. Here too the lending agencies are hesitant to advance any credit facilities for long-term investments which can make land use more sustainable. There is evidence that this has resulted in the neglect over 100,000 ha of developable land by private owners.

Another land tenure issue related to land degradation is the fragmentation of agricultural lands. Farm sizes in Sri Lanka are very small and this is particularly so in the hill country and the wet zone. It has been estimated that nearly 80 per cent of our farms are less than 1.2 ha. and over 40 per cent of them are below 0.4 ha. Small farms apart from their productivity implications, tend to present problems for soil conservation in particular and watershed protection in general. In this set-up even if some farmers adapt soil conservation measures, they will have to suffer if their neighbours on upper slopes and up stream areas do not follow the same. Therefore people's participation becomes absolutely necessary in handling problems of land degradation under small farm conditions.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Immediate action aimed at bringing LDO permits closer to free titles on an experimental basis.	Immediate	100,000	M/LI&MD, Land Commissioner
2 Introduce a strategy for encouraging voluntary land consolidation as a solution to land fragmentation.	Medium	250,000	Commissioner of Agrarian Services, NGOO
3 On short term basis organise farmers in critical catchment areas as watershed farmer organisations appear to be a promising institutional approach for soil conservation on highly fragmented farmlands.	Medium	200,000	NADSA, NGOs
4 Develop a mechanism to induce commercial banks to accept LDO land as security on credit for land development.	Immediate	100,000	Central Bank

6

PROBLEMS IN LANDUSE PLANNING

BACKGROUND:

Land use planning in Sri Lanka is as old as her history and is evident in the early settlements and agriculture in the purana villages based on cascading systems of tanks as well as in the landscape gardens of Sigiriya. However, the beginning of modern landuse planning could be traced back to the provision for 'Mapping Out' of State lands in the Land Development Ordinance of 1935. Today, while Land Use Planning functions are scattered among several State agencies the primary responsibility rests with the Land Use Policy Planning Division (LUPPD) of the M/LI&MD which performs a key role at the national and regional levels.

Land use planning activities of the LUPPD are now decentralized to districts each having a land use planner. Some efforts are being made to organize land use planning activities at the Provincial as well as AGA/DS levels. However, lack of an overall landuse strategy at the national level tends to create difficulties of integrating such activities within a broader framework. This also creates difficulties in integrating the work of different line agencies which tend to have their own approach to their landuse problems.

The formulation and implementation of a National Land Use Plan is an ideal answer although it is not a quick or easy task. In order to plan at the National level, there should be a constant flow of information from the village and sub-regional level so that a 'top-down' approach may be avoided. At the upper end it needs a visionary approach which takes into account a variety of future scenarios as well as the overall national development strategy with which it must be compatible. It should also have firm political backing so that its implementation will not be hindered.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Establish a National Land Use Plan Project that functions within the LUPPD or the Technical Secretariat to the proposed National Land Commission.	Immediate	100,000	M/LI&MD, LUPPD
2 Implementation of the National Land Use Plan.	Medium and Long-term	2,000,000	LUPPD, PCs

7

LAND AT HIGH ELEVATIONS (ABOVE 1500 METRES)

BACKGROUND:

The need for protection of land above 1525 metres (5000 ft) was recognized and adopted as a policy by the British in the last century. This policy remained unchanged despite frequent infringements by encroachers and even by State organizations. The pattern of land use above this elevational limit today indicate the haphazard nature of land allocation and use over the years. The time has come to review this policy, taking into consideration both conservation needs as well as development priorities in the areas.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Map out existing land use above an elevation of 1500 metres at an appropriate scale, (viz 1:10,000). Formulate an indicative land use map for areas above 1500 metres.	Medium	25,000	LUPPD MASL (Env't Unit)
2 Preparation of the final land use plan and its implementation with the co-operation of the provincial and district administration.	Medium	1,000,000	LUPPD, PC, DEA/Nuwara Eliya

8

NATURAL HAZARD MANAGEMENT

BACKGROUND:

Although relatively free from extreme geophysical events, Sri Lanka is subject to a variety of natural hazards such as landslides, floods and droughts. In addition to the economic cost of these events which may be of the order of several hundred million rupees every year, there are great social costs.

The amount of funds needed to rehabilitate those affected by natural disasters are increasing rapidly. The tropical depressions, though not recurring as frequently as droughts and floods, can have a devastating impact particularly in the East and in the North, as witnessed in 1979. Coastal erosion, particularly on the West coast, is a more frequent occurrence and although its impacts are not so dramatic and the number of people affected at a given location may be smaller, its cumulative impact over the years is very considerable and is rising.

The existing arrangements for natural hazard management are essentially based on 'fire-brigade' type short-term approaches. Social service authorities and the local administration take the responsibility of rushing relief supplies only after the disaster had struck. Long-term planning aimed at minimising the risks involved are rare. There is a need therefore, to promote appropriate human adjustments to each type of hazard in different areas.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Establishment of a Natural Disaster Relief Fund, and ensuring its efficient management from the long-term perspective.	Medium	2,500,000	M/SS, M/E&PA
2 Extension of the on-going Landslide and Hazard Mapping Project by the NBRO to Kegalle District.	Immediate	1,000,000	NBRO
3 Development of long-term plans and institutional mechanisms for rehabilitation of people affected by natural disasters.	Medium	2,500,000	M/SS, M/LI&MD

9

MULTIPLICITY OF INSTITUTIONS AND WEAK ENFORCEMENT OF LEGISLATION**BACKGROUND:**

The subject of 'Land' falls within the purview of a large number and variety of Government agencies and Provincial Councils. Over 60 agencies and some fifteen Cabinet Ministries have responsibilities of some sort or other related to land. This dispersion of functions is partly inherited from the colonial past but is mostly due to an unco-ordinated multiplication of State agencies since National Independence. The effect had been inefficient development, management and conservation of land resources.

The Land Commission (1985) which drew attention to the problem of institutional multiplicity in the land sector, reported that there is a real need for rationalization of the functions of existing agencies in order to optimize the use of available resources even if it results in the re-organization and merger of some of them. The inertia resulting from long periods of existence has to be overcome effectively and re-organization of the network of institutions concerned with land matters should be undertaken with least hardship to institutions and employees.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Launch a study which would examine from a long-term perspective, the issue of multiplicity of institutions in the land sector.	Immediate	50,000	SLIDA, Universities
2 Promote efforts at re-organizing and effecting better co-ordination of existing institutions.	Immediate	250,000	M/PP&I, M/E&PA
3 Enhance the institutional capability in the area of watershed management incorporating several aspects of land degradation.	Medium	1,000,000	MASL (Env't Unit)

WATER RESOURCES

OVERVIEW

During the early period of Sri Lanka's hydraulic civilization, the population was confined mainly to the Dry Zone and the hill country was largely preserved as a watershed. This helped to regulate the flow of streams, the majority of which drained from the hill country. During the colonial era, with the opening of hill country areas for plantations, the forest cover was gradually and steadily removed, resulting in heavy soil erosion, siltation of waterways and floods during the rainy months.

The extensive irrigation network which flourished during the hydraulic civilization collapsed around the 12th Century due to both environmental as man-made reasons. Emphasis on irrigation development regained only since the 18th Century.

Today, irrigation absorbs about 10 per cent of public sector investments and 22 per cent of agricultural lands are under irrigation.

The emerging problems relating to water at present is the depletion in the quantity available, deterioration in quality and the emergence of health issues. The only area with a water surplus is the Western Wet Zone while the most acute deficit exists in the northern, north-western, north-eastern and south-eastern zones. The available supplies are threatened by pollution and the efficiency of utilization is sub-optimal due to poor management. Most of the surface water potentials have already been exploited and the groundwater extraction remains yet to be fully tapped. The pollution threat of groundwater resources is rapidly becoming a critical issue.

Depletion of water quality and problems concerning watershed management are highlighted at several forums including the NESC. On a directive from the NESC, action has been initiated by the M/LI&MD to prepare a Water Resources Master Plan.

Principle environmental issues to be addressed are:

- (1) Water management, including conservation;
- (2) Alleviation of water quality; and
- (3) Prevention of adverse effects due to irrigation.

1

WATER MANAGEMENT AND CONSERVATION**BACKGROUND:**

The demand for water is increasing mainly due to population expansion and the development and extension of uses of this resource. The main uses are for irrigation and energy generation.

Today, about 22 per cent of the agricultural land is under irrigation. Over 75 per cent of the paddy from which rice, the staple diet of the Sri Lankans, produced is under irrigation.

In spite of several irrigation development programmes during the last two decades, the cropping intensity of the irrigable land under major schemes has been stable around 140 per cent intensity while that for minor schemes had been around 90 per cent. The main reasons for this stagnation is due to inefficient water management within schemes and disparity in income from the crops cultivated. For example the water use rate in paddy cultivation has remained for the last two decades unchanged around 15 acre-feet of water per acre of paddy although the crop can be produced with as little as 4-5 ac-ft of water. It is now realized that the desired reduction may only be achieved through active participation of the users.

Success in saving water and reduction of adverse impacts due to irrigation via efficient water management in irrigation projects where farmer organizations (FOO) have been promoted on a pilot scale supports the extensive adoption of such a participatory approach in water management. As a result of water saved from the head areas and by more equitable water distribution within the schemes managed in association with FOO, large extent of lands in the tail reaches of those schemes can be brought under cultivation. Furthermore, as a direct implication of FOO, the understanding between officers and farmers and among the community have improved, in the benefit to the farmers.

Preparation and implementation of a comprehensive Master Plan addressing the present and future environmental issues relating to water resources utilization should be carried out on a priority basis.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Water Resources Master Plan to be prepared and implemented.	Immediate	150,000	M/LI&MD
2 Undertake awareness and educational programmes on water management measures for officers and farmers and impart necessary training to the officers working with FOO. Establish appropriate units within the Provincial Councils and AGA offices in support of the monitoring programme proposed above.	Immediate	200,000	PCs, AGA/DSs
3 Establish and promote FOO covering about the 22,000 irrigation schemes. Implement the participatory management policy of the Government and establish the basic institutional facilities to support the activities of FOO. Ideally the grassroots level NGOs should be utilized towards formation of FOO.	Medium to Long-term	250,000	Commissioner of Agrarian Services, NGOs
4 Research should be carried out to address issues in the establishment of FOO.	Immediate	200,000	IMPISA, Universities

2. ALLEVIATION OF WATER QUALITY PROBLEMS

BACKGROUND:

Studies indicate that the quality of water in some important rivers such as Mahaweli and Kelani is deteriorating. The main pollutants are chemical residues, fertilizer leachate and silt loads—all of which originate mainly from poorly managed agricultural lands. However, there is no continuous programme for monitoring the level of pollution in most of the important waterways, due to lack of technical facilities and expertise and, deficiencies in the institutional system. Furthermore, data already collected through various ad-hoc studies have not been fully analyzed and information is not available to decision-makers for similar reasons.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 i Monitoring in a continuous basis of the nature and extent of on-going agro-chemical pollution of selected rivers and other waterbodies of the country;	Immediate	250,000	FOO, Commissioner of Agrarian Services
ii Progressive development of the institutional system;	Medium to Long-term	100,000	IMPSA, Com of Agrarian Services
iii Training of officers and farmers to carry out the water management task;	Medium	100,000	- do -
iv Reporting of data on the nature of pollution and its causes;	Medium	50,000	- do -
v Undertaking of research studies as required;	Immediate	100,000	Universities, IIMI
vi Education of the public on the adverse effects of pollutant build up in water.	Immediate	100,000	CEA, NIE

3

PREVENTION OF ADVERSE ENVIRONMENTAL IMPACTS DUE TO IRRIGATION**BACKGROUND:**

The evidence of adverse environmental effects due to poor irrigation is mounting. Among the more important effects are development of salinity, water logging, siltation, increase in the disease transmitting vectors and water-borne diseases, loss of land due to borrow areas and ecological problems related to large inland water bodies. Unfortunately, the extent of damage to the environment and the scale of multiplication of these problems are not known to any degree of accuracy. For instance, the extent of the land area affected by soil salinity in premier irrigation schemes, or the extent of land area which has become barren or been abandoned due to earth collection or to the spread of malarial and other parasitic diseases within the country are not adequately known. It is therefore, necessary that baseline data with respect to these problems is established and monitored regularly.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Assess the nature of environmental problems through a study covering several important irrigation schemes of the country; Verify the extent of damage and the scale of degradation of the environment; Establish the baseline status with respect to priority environmental issues;	Immediate	400,000	Universities
2 Undertake a study to: a Identify environmental parameters for monitoring, b Determine methods to monitor; Design a strategy to monitor the above environmental parameters;	Medium to Long-term	250,000	Universities, NBRO
3 Train and educate people, especially members of FOO on the adverse effects and ways of alleviating problems; Train members of FOO to be vigilant on environmental problems and support them to conduct awareness programmes; and	Medium to Long-term	100,000	CEA, NIE
4 Launch research studies to alleviate adverse effects and to strengthen beneficial environmental effects.	Immediate	100,000	Universities

MINERAL RESOURCES

OVERVIEW

Sri Lanka is historically renowned for the quality of some of its endowed mineral wealth. Recent trends in the demand for minerals both locally and from outside have increased the activity both in the traditional and the new. This has led to increased exploitation of these non-renewable resource with little regard to either the management of resources and the protection of the environment. In certain sectors activity has proliferated beyond controllability and is carried on illicitly. In illicit gemming and river dredging for gems the damages are severe. On the other hand, authorized activity is largely unmonitored resulting in similar degradation of land as a resource, and often linked with other environmentally damaging consequences such as pollution of water bodies, spread of diseases and damages to eco-systems. Furthermore, it leads to mismanagement of the resource itself having a long term effect on the quality of life as a whole.

Inadequacies in the regulatory system, in servicing the industry and in research are apparent.

Main issues discussed here are:

- (1) Existing legislation on mineral resources does not address environmental protection/issues adequately.
- (2) Weak institutional structure for servicing the mineral sector in assisting, monitoring and guiding mineral industry.
- (3) No comprehensive data base giving Mineral Information for the industry.
- (4) No environmental assessment is done in mineral based activities.
- (5) Inadequate research in mineral based activities.

1	EXISTING LEGISLATION ON MINERAL RESOURCES DOES NOT ADDRESS ENVIRONMENTAL PROTECTION/ISSUES ADEQUATELY			
BACKGROUND: The Mines and Minerals Law No.4 of 1973 is being amended but excludes dealing with Radio Active Minerals, Gems and Hydrocarbons. General conditions of 'mining' or related activities like 'land filling' that can create serious environmental problems are not correct.				
RECOMMENDED ACTION		IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
Review amend or change existing laws to accommodate environmental considerations. Pertaining to the particular mineral activity		Immediate	20,000	Ministry in-charge of the project of mineral resources

2	WEAK INSTITUTIONAL STRUCTURE FOR SERVICING THE MINERAL SECTOR IN ASSISTING, MONITORING AND GUIDING MINERAL INDUSTRY			
BACKGROUND: The present Geological Survey Department is inadequately staffed and the Mines division under the Director has completely deteriorated.				
	RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
	Strengthen and improve the GSD and its Mines and Minerals branch in line with the ADB report.	Medium	200,000	Ministry incharge of the subject of Mineral Resources.

3	NO COMPREHENSIVE DATA BASE GIVING MINERAL INFORMATION FOR THE INDUSTRY			
BACKGROUND: There is no comprehensive Mineral data base although a mineral map and the publication of Economic Minerals of Sri Lanka exist. A wealth of unpublished information exist.				
RECOMMENDED ACTION		IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
Collate and compile a comprehensive Mineral information data base and keep updated.		Immediate	100,000	GSD

4	NO ENVIRONMENTAL ASSESSMENT IS DONE IN MINERAL BASED ACTIVITIES		
BACKGROUND:			
No environmental assessment is being done on specific minerals or project sites leading to lack of appreciation of sustainability of the environment.			
RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
Undertake an assessment of currently exploited mineral species. Prepare guidelines, issue & coordinate monitoring action with peripheral authorities. Areas that may be covered: clay, gems, dredging, beach sands, quartz mica, felspar, dolomite, shell/coral, river sand, rock, quarrying of earth fill.	Immediate	450,000	GSD

5	INADEQUATE RESEARCH IN MINERAL BASED ACTIVITIES			
BACKGROUND: There is inadequate research into the resource utilization and mining, quarrying or extraction methods currently seen in Sri Lanka. Unseparation mining is prevalent specially in brick and the clay extraction, extraction of Topaz, etc.				
RECOMMENDED ACTION		IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Publicize, offer and promote research: i Gem mining and river bed dredging for gems; ii Exploitation of clay fields; iii Vein Quartz, Felspar, felspar mining; iv Mica, graphite mining; v River sand, dredging; vi Ornamental rock quarrying.		Medium	500,000	M/E&PA, CEA, GSD

6

GEM MINING

BACKGROUND:

Long before Ceylon tea gained its international recognition, Ceylon gemstones were known for their unique quality and variety. To date, they have been Sri Lanka's most important mineral resource. Of the 5 percent mineral export earnings (1989), over 80 percent originated from gemstones. While around 15,000 gem pits operate legally, it is estimated that an equal number are illegal. The State Gem Corporation (SGC) was established in 1971 to supervise this industry, but it has not had sufficiently trained staff to enforce existing regulations.

Most of the gemstone deposits on the Island lie within a geologically narrow zone (240 by 64 square km) with Ratnapura as its leading center. Recent geological surveys reveal that the potential for gemstones is about 50 percent higher than estimated earlier. Most gemstone mining follows the traditional, small-scale pit mining with lateral tunnelling. Mechanized surface mining has been carried-out since 1985 in the North Central and Southern Province.

There are many negative environmental effects of gem mining. A detailed study in Ratnapura District indicated that damages from traditional methods included: soil erosion, land slides, depletion of plant cover, sedimentation of streams and rivers, effect on soil profiles (loss of fertile layer), groundwater depletion and pollution, surface water pollution, outbreak of vector-borne diseases such as malaria, destruction of river beds, and loss of arable lands. In addition to the above, damages from mechanized practices included: destruction of landscape (and effect on tourism), destruction of forest cover, impact on wildlife, and disturbance of groundwater table and aquaducts by extensive excavations.

The proponents of mechanized gem mining argue that it is a significantly more efficient method of gem exploitation that potentially lends itself to more effective enforcement of environmental guidelines than traditional small pit mining. Others argue that traditional methods -- under proper supervision -- provide Sri Lankans with jobs (135,000 employed in 1988) and income. This debate can only be resolved through evaluation of past practices and a review of what can be instituted through bolstered agency resources, more rigorous enforcement of mining regulations, and a solid partnership among Government, miners, gem traders and local communities.

NATIONAL ENVIRONMENTAL ACTION PLAN

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Strengthen the capacity of the SGC or successor organization to carry out its environmental responsibilities more effectively. To do this, the following steps should be considered. i Establish an Environmental Cell under the Chief Executive of the responsible GOSL gem mining agency to manage all environmental matters related to gem mining. The unit should develop guidelines to be followed by all those involved in gem mining activities from start to finish. It should also conduct a series of awareness programs in the community affected by the gem mining activities. ii The staff of the agency should be strengthened in number, and given enhanced authority, training and motivation to enforce the relevant environmental provisions of the law more effectively. Staff facilities should also be expanded. iii The link between the agency and the CEA should be strengthened. The agency should be required to submit quarterly status reports on its on-going gem mining operations, reclamation plans and progress of reclamation of abandoned sites to the CEA. iv The agency's enforcement officers' role should be re-oriented to include environmental monitoring which should be reported in the quarterly submission to the CEA. They should also become members of the District and/or Provincial Environmental Councils and serve as the link between the gem mining agency and others at the local level.	Immediate	300,000	

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
2 Integrate the location of gem sources with future land use plans prepared for gem mining areas. A detailed study should also be initiated on methods for reclaiming mined areas. 3 The financial bond deposit for deep mining projects should be increased substantially, and refund issued only after satisfactory rehabilitation is completed. Incentives should be developed to encourage payment of the financial deposit – possibly through streamlined approval and permit issuance system. Disincentives for operating illegal mines should be increased. 4 The pros and cons of mechanized versus traditional pit mining should be evaluated in the context of Sri Lanka's high man-land ratio, the need for foreign exchange, and the need to maintain an environmental balance. If mechanized mining is to proceed, Environmental Impact Assessments should be mandatory. Similarly, EIAs should be required for all areas extending beyond ten acres when auctioned for traditional mining practices. Studies of the cumulative impacts of small, traditional pit mining should be developed for comparative purposes. 5 In addition to gem mining, Government - namely CEA, in collaboration with the M/IS&T and other agencies – should assess the extent to which the mining sector (and the draft Mining Law) that includes, inter alia, clay pits, limestone quarries, kaolin pits and mineral sands, is provided with appropriate environmental guidelines.	Medium	100,000	

COASTAL RESOURCES

OVERVIEW

Sri Lanka's coastal region along its 1,585 km of coastline supports fisheries that supply the population with most of its animal protein, recreation at resorts and national parks that draw most of the country's tourists, as well as industry, aquaculture, agriculture, and mineral sand extraction. Over the past few decades population increases and coastal development have become particularly intense in the Wet Zone. With passage of the Coast Conservation Act of 1981 a narrow coastal zone is subject to regulation by the Coast Conservation Department (CCD), which evaluates and must grant permits for major development projects, and prepares and implements a CZMP, in close co-operation with other Government agencies such as the NARA, the UDA, GCEC and CEA.

Environmental issues of concern to the Government and the general public include degradation of natural habitats (mangroves, estuaries, lagoons, coral reefs), water pollution, coastal erosion and accretion from natural and human causes (such as sand mining), and the future impacts of rising sea levels on infrastructure. The CCD has focused attention on most of these issues, and it has obtained substantial donor support for a wide range of coastal resource management activities, including institutional strengthening, planning, erosion control, and research. Given the coastal orientation of many proposed national development plans (viz., Galle Port development, Koggala Investment Promotion Zone, accelerated development of Southern Province and proposed Muthurajawela development), continued coastal protection initiatives within the framework of the Coastal Zone Management Plan will be essential.

Issues resulting from environmental impacts on coastal resources can be prioritized as follows:

- (1) Impacts of coral mining;
- (2) Impacts of sea level rise;
- (3) Pollution of coastal areas;
- (4) Effect on persons living on erosion-prone coastal areas.

1	CONTROL OF CORAL MINING		
BACKGROUND:			
The coral reefs along Sri Lanka's South Western coasts had been subject to large scale destruction from mining of coral for the lime required in the building industry. This activity is particularly intense in Akurala, Seenigama, Kahawa, Habaraduwa, Madiha and in the Rekawa village. Mining reduces the effectiveness of coral reefs in acting as a barrier against sea erosion, destroys habitat of a large number of marine fauna and flora including several economically important fish species thus considerably reducing biodiversity and leads to declined potential for tourism. Coral renew at a very slow rate and current patterns of its exploitation is of serious concern, demand for coral is likely to grow in the future, control of coral reef destruction is urgent. It has been difficult to enforce the legislation which prohibits coral mining and related activities mainly because of the significant socio-economic dislocations caused by the prohibition of these activities. Although attempts have been made in the past to provide alternate employment and land to those engage in coral mining these efforts have had limited success, mainly because such programmes were not supported by a strong planning base or reliable information on socio-economic data. The development of alternative sources is also necessary to ensure that adequate quantities of lime to supply the construction industry is available. One of the major alternative sources identified are Dolomite found in the Central Province.			
RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Implement a planned programme to provide alternate employment and land to those engaged in coral mining in the Southern province.	Immediate	100,000	Southern Provincial Council
2 Implement a programme to develop dolomite resources as a source of lime required for the construction industry, and to make it available at a competitive price.	Immediate	200,000	M/H&C

2

SEA LEVEL RISE

BACKGROUND:

Impacts of possible sea level rise is of major concern in view of the high population density and intensity of development in the coastal areas, especially the western and southwestern coasts. Inundation, erosion, flooding of coastal areas, and salt-water intrusion and other impacts on water quality would be of particular concern.

A good part of Sri Lanka's coastline is backed by estuaries, lagoons and other wetlands. Inundation would, therefore, be a major impact of sea level rise. Sri Lanka's coastal fisheries are dependent upon the productivity of coastal wetlands. Changes in extents of wetlands and changes in the salinity regime of estuaries would significantly affect coastal fisheries. Inundation would also pose a considerable threat to infrastructure and communication facilities located along the coast. Damage caused by inundation to tourism facilities would be considerable.

It has been shown that, on an average, a one metre rise in sea level would cause shores to erode 50-200 metres on the sandy beaches, even if the visible portion of the beach is fairly steep. Even now, coastal erosion is a critical problem in certain areas. It will become much more serious and widespread in the event of accelerated sea level rise, and protective structures now in place will be rendered ineffective.

Salt water intrusion is of particular concern in Sri Lanka because the intake for the supply of water to the City of Colombo is located only 7 km upstream in the Kelani Ganga. Salt water intrusion will also result in increased salinity of groundwater, impaired effectiveness of septic tanks, and adverse effects on coastal agriculture.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Prepare an initial report which identifies: (a) the magnitude of the problem; (b) potential site specific impacts and particularly vulnerable areas; (c) priority research and information needs; and (d) recommendations on optimal response strategies.	Immediate	75,000	CCD, M/E&PA
2 Conduct priority research activities investigations and surveys on critical sites.	Immediate	500,000	CCD, M/E&PA Universities
3 Prepare a strategy for minimising adverse impacts of sea level rise.	Medium	75,000	CCD, M/E&PA

3

COASTAL POLLUTION

BACKGROUND:

Coastal pollution is a growing problem in urban areas. Estuaries and lagoons already seriously affected by pollution include Negombo Lagoon, Kelani estuary, Lunawa Lagoon, Walawe estuary, and the Batticaloa Lagoon. In a few cases the cause is industrial alone, but in others impacts result from combined industrial effluent, domestic/municipal sewage, and agricultural wastes. In key tourist centers like Hikkaduwa and Bentota, sewage disposal is a problem faced by most hotels, all of which use septic tank systems, and only a few hotels have treatment plants. Potential health hazards to residents as well as tourists, and longer term degradation of coastal waters, are likely to become increasingly serious economic and environmental threats. The problem is recognized by the CCD, CEA, NARA and the Tourist Board, and local authorities. In some places the remedy is central sewage treatment plants for groups of hotels. Overall, however, little can be done to attack the problem effectively and efficiently without careful water quality monitoring and assessments of the pollution sources, quantification of the risks to health and environment, and evaluation of the costs and benefits of various control strategies.

Given the importance of tourism to Sri Lanka's economy, the local importance of coastal fisheries and the vulnerability of both to the effects of water pollution, every effort must be made to anticipate and prevent significant pollution damage from domestic, industrial, and tourist hotel sources.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Conduct feasibility studies for central sewage treatment/disposal facilities in key tourist development areas along the coast.	Immediate	100,000	CEA
2 Establish water quality monitoring and risk assessment programmes in critical estuary/lagoon systems and develop cost effective recommendations for anticipating and preventing adverse water pollution and solid waste disposal impacts on human health and the environment.	Medium	100,000	CEA

4

RELOCATION OF PERSONS AWAY FROM EROSION PRONE COASTAL AREAS

BACKGROUND:

Increasing population has led to a greater density of housing in the coastal zone. Some of these settlements are located in erosion prone (eg. Negomebo, Dehiwela, Mt Lavinia, Lunawa, Moratuwa, Akurala, Weligama, Kandakuliya) or otherwise vulnerable areas. Some of these fragile stretches of the coast are heavily populated with people living in houses mainly of temporary structures, built almost at the edge of the coast. Many of these temporary structures are built on government land. These allotments are small, the land infertile and the population often poor, and thus compelled to exploit available natural resources such as coral and sand. Sanitation is poor, supplies of potable water and the living conditions are below acceptable levels.

Relocation of these communities have to receive serious consideration because:

- It is inadvisable to construct permanent houses in these locations with sanitation, water and other community facilities since they are erosion prone;
- In case of natural disasters, large sums have to be spent on relief and relocation of those affected often unplanned and under emergency situations;
- Their continuance in these locations tends to lead to accelerate the degradation of the coast.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Conduct a survey to obtain information on: (a) the nature and magnitude of the problem; (b) to identify the more critical areas where re-location needs to be undertaken on a priority basis on; (c) identify suitable alternative lands and infrastructures facilities required;	Immediate	300,000	CCD, Universities
2 Implementation of relocation programme;	Medium	1,000,000	M/H&C and Land Commissioner
3 Implementation of a programme in one or two areas, including establishment of green belts in the vacted coastal areas.	Medium	250,000	NGOO

FOREST RESOURCES

OVERVIEW

From ancient times, the natural forests of Sri Lanka were carefully nurtured and preserved. Their importance in proper land use, especially with regard to agriculture, was well understood. Forests were only minimally disturbed and then, too, only to provide simple and basic human needs. As Nanayakkara (1987) observed: *"Man at that time achieved self-sufficiency in food, practised agro-forestry in homesteads and also lived in harmony with a neighbouring forest environment. There appeared to be a close inter-relationship between man and forest, man depending on forest for a variety of forest products and medicines."*

Unfortunately, the dense forests did not remain so for much longer. The exploitation of forest resources which increased under the Portuguese and the Dutch, was further intensified under British rule. The Forest Department, set up in 1887, did not emphasize its conservation role until three decades ago, since its original mandate was to supervise timber extraction for the colonial power and to prevent the local population from doing so.

Deforestation in Sri Lanka occurred at a rapid rate between 1880-1990. Starting at about 84 per cent of the land area in 1881, forest cover dropped to 70 per cent by 1900, 44 per cent by 1956, and an estimated 24 per cent (or 1.58 million ha.) of the land area remained under natural forest by 1989. The consequence of this alarming rate of deforestation (meaning complete clearance of existing forests and their replacement by other forms of land use) was not only the loss of the island's natural scenic beauty. The most important result is that deforestation is linked with the loss of species and genetic diversity which, from an international perspective, is perhaps the most important environmental issue in Sri Lanka. Of over 3,600 species of flowering plant species, 24 per cent are endemic. Ninety-four percent of the flowering plant endemics inhabit the Wet and Intermediate Zones, while the remaining 6 per cent are found in the Dry Zone.

Other reasons for the importance of forest cover for Sri Lanka go beyond the mere 1.7 per cent of GDP that the forest sector supposedly contributes to the economy in direct value terms. Forests regulate stream flows, stabilize steep land, control soil erosion, reduce flood intensities, prevent land desiccation, and protect agricultural lands and human settlements. Natural forest products such as berries, fruits, nuts, game, and other products serve local economic needs even if they never reach urban or foreign markets.

The issue of deforestation and forest degradation (caused by over exploitation of forests for timber and fuelwood, illegal felling, over grazing, and illicit gem-stone mining) is being addressed by a large number of Government agencies, donors, and NGOs. While not without its share of controversy, the issuance of the Forestry Master Plan in 1986 resulted in the development of a common body of information and analysis and broader public participation in a forestry development programme over a five-year period (mid 1990 to 1995). Support for this programme is expected through development grants and loans from the World Bank (IDA), British ODA, FINNIDA, UNDP, ADB and the NORAD.

However, there are some issues remaining in the existing approaches to reduce Sri Lanka's deforestation problem. These are namely:

- (1) Lack of community development programmes based on forestry;
- (2) Inadequacy of institutional capacity for forest protection;
- (3) Inadequate capacity for timber utilization research.

1

LACK OF COMMUNITY DEVELOPMENT PROGRAMMES BASED ON FORESTRY**BACKGROUND:**

The average forest cover in Sri Lanka in 1989 has been estimated at 24 per cent of the land area of the country, but the forest cover in the Wet Zone where the forests exhibit the highest degree of species diversity and endemism is only around 8 per cent of the land area of that region. The value of the natural forests in the Wet Zone has been widely recognized, in addition to the importance of these forests as genetic reservoirs, they also play a very crucial role in watershed protection. Therefore undoubtedly all the remaining areas of natural forests in the Wet Zone should be strictly protected and not be allowed for further degradation.

However, environmental considerations cannot be divorced from social conditions, for the causes of encroachment and over exploitation are usually social and economic. The forest development needs, therefore to be planned in ways which bring benefit to the people around the forest and which reduce the risk of damage to the forest resource. Such benefits can be provided by maintaining buffer zones around the forest which are used to provide limited economic activities associated with the forest. These activities include employment, collection of small timber for domestic purpose and collection of forest produce other than timber etc. The forest conservation and peoples participation in such a programme could be achieved through a community development programme which is aimed at strengthening of the villager's social and economic capacity.

These programme should be aimed at achieving following objectives : (a) Strengthen villager's capacity for community action; (b) facilitate villager's access to government incentives and programmes; (c) promote land restoration in the forest buffer areas, both State and private lands; (d) facilitate acquisition of land tenure security for the villagers; (e) create village nurseries and demonstration plots; (f) training and transfer of technical know-how to villagers; (g) facilitate village action for eco-restoration of degraded lands; (h) facilitate and encourage village action for development of non-farm and non-forestry related income generation activities and marketing and credit facilities.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Develop and implement a community development programme emphasizing forestry in a few demonstration locations adjacent to natural forests in conservation areas. Although such a programme should be supported by the Forest Department, local bodies and NGO at district level should provide co-ordination and direct links with people.	Immediate	25,000	FD, NGO

2

INADEQUACY OF INSTITUTIONAL CAPACITY FOR FOREST PROTECTION**BACKGROUND:**

Illegal felling within state forests for timber and fuelwood is common in Sri Lanka. Illegal or unrecorded cuttings amounted 195,000 cubic meters (m³) of sawlogs in 1984 and 2.0 million m³ of firewood on an annual basis. This activity has severely depleted the natural and plantation forests of its valuable timber species. Teak and Mahogany plantations have been badly affected because of these illegal cuttings. Additionally, large extents of state forests are encroached upon for settlements and agricultural development. According to the 1979 Land Encroachment Survey, nearly six percent of Sri Lanka's land area was under encroachments, but much of this may have been on road reservations and other non-forested land. Recorded forest offenses for illicit tree felling, transport and encroachments amounted to 4,400 in 1989.

Forest fires have also been an important cause of forest destruction. According to Forest Department records, an average of 1,500 ha. of forest plantations are damaged every year by fires.

Efforts to combat illegal felling, encroachments and forest fires have been limited in the past due to the shortage of manpower and financial resources in the Forest Department. Nearly 50 per cent of the time of the Beat Forest Officers (BFOs) and Range Forest Officers (RFOs) is spent on enforcement of the Forest Ordinance, with little time for actual forest management and silvicultural work. At present, each BFO is responsible for protecting nearly 5,000 ha. of forests and forest plantations, which is a virtually impossible task. The Forest Department should be provided with sufficient mobility and staff for efficient forest protection and the vacancies of the territorial staff of the Forest Department should be filled immediately.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Establish a special Forest Protection Division in the Forest Department to combat illegal fellings, encroachments and forest fires. This Forest Protection Division should be provided with sufficient mobility and staff. Staff requirements: Deputy Conservator of Forests - 1 Legal Officer - 1 Asst. Conservator of Forests - 3 Foresters - 29 Range Forest Officers - 23 Beat Forest Officers - 42	Immediate	30,000	FD
2 Revise the existing arrangements of Forest Ranges and Beat based on the availability and distribution of Natural Forests and Forest Plantations in the area. This includes recruitment of staff to new Forest Ranges and Beats. (Approximately 25 per cent increase of Range and 50 per cent increase of Beats are recommended).	Immediate	15,000	FD
3 Improve mobility of territorial staff of the Forest Department by providing Motor Cycles.	Immediate	12,500	FD

3

INADEQUATE CAPACITY FOR TIMBER UTILIZATION RESEARCH

BACKGROUND:

The timber utilization Research Branch which was with the Forest Department was handed over to State Timber Corporation in 1986 with a view to streamlining the research related to utilization of timber. But due to lack of facilities and trained staff in this branch the progress of research during the recent years has been poor. The technical information on lesser known, non-traditional species is not available and this information is very vital in order to popularize the use of these species. Therefore, timber utilization research should be strengthened and the research activities should be focused on the use of non-traditional species and rational utilization of timber.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Timber Utilization Research Branch should be strengthened by providing necessary staff and equipment and preferably relocated in the FD. Staff requirement: Research officers - 3 Technical officers - 4	Medium	35,000	M/LI&MD, FD, STC

BIODIVERSITY AND WILDLIFE

OVERVIEW

Sri Lanka has more endemic species per unit area than any other country in Asia, and is one of eleven areas in the tropics identified by the Committee on Research Priorities in Tropical Biology (NAS, 1980) for special attention because of its high levels of biological diversity, endemism and forest conversion to other uses. Sri Lanka's biodiversity, particularly the wildlife, attracts tourists and is of economic significance.

Two factors pose serious threats to the preservation of species diversity in Sri Lanka: (i) deforestation and forest degradation particularly in the Wet Zone; and (ii) selective exploitation and promotion of economically valuable species, such as elephants, certain timber species (calamander, satinwood), and certain medicinal and ornamental plants. Damming and filling of wetlands and destruction of coral reefs pose additional threats to species preservation. Nearly 16 per cent of flowering plants and 28 per cent of ferns and fern allies, including 12 endemic genera, are considered threatened.

Over 15,000 km² are under the jurisdiction of the Department of Wildlife Conservation (DWLC) or Department of Forestry (FD), and Sri Lanka is one of a handful of countries that have allocated more than 10 per cent of land area to wildlife protection. This represents a significant commitment to conservation of biological diversity and wildlife protection. The size and distribution of Sri Lanka's wild elephant population is testimony to the success of Sri Lanka's past wildlife conservation efforts. However, there is significant encroachment on protected areas and wildlife habitat, and DWLC and FD are short of budget and staff for conservation activities.

The principal conservation issues to be addressed under the EAP are as follows:

- (1) Inadequate coverage and management of existing protected areas. The distribution and fragmentation of existing protected areas raise questions regarding the effectiveness of the protected area network in species preservation. Despite the concentration of biological diversity (94 per cent of endemic plants and 75 per cent of endemic fauna) in the Wet Zone, 90 per cent of protected land area is located outside the Wet Zone. The 40 per cent of protected areas of less than 1000 hectares are nearly all "biological islands" surrounded by agricultural and other developments, and are believed to fall far short of the minimum critical size for effective preservation of diversity. Moreover, except for a few Wet Zone forest reserves and wildlife protection areas under the Mahaweli Authority, protected area management plans are lacking and institutional capacities of DWLC and FD to implement them are inadequate.

- (2) Mobilization of resources to support conservation. Although there is significant interest in support of conservation activities on the part of foreign donors, NGOs and, potentially, the private sector, mechanisms to mobilize resources for support of conservation activities are lacking.
- (3) Wetlands and wild elephants are significant resources under particularly intense pressure that need new and immediate as well as long term attention. Existing programmes must be substantially augmented to address imminent and potentially irreversible losses.

A more detailed description of these issues and proposed actions follows.

BACKGROUND:

Ninety per cent of protected areas (nature reserves and sanctuaries) are concentrated in the Dry Zone and exhibit low levels of biodiversity and endemism relative to the Wet Zone. Most Dry Zone protected areas, with the exception of new areas created since 1980 under the Accelerated Mahaweli Project, were created as hunting reserves, and therefore do not necessarily reflect biodiversity values. Only 2.6 per cent of the Wet Zone is under protected status, and this represents far from adequate coverage of the high biological diversity and endemism levels there. Thirteen of the eighteen reserves in the Wet Zone occur in small, fragmented patches of less than 1000 ha. Because of the small size of these "biological islands," which are subject to intense population pressure, the sustainability of their species preservation capacity is questionable. Although the total acreage under protected status has increased, there are only two reserves that contain 100,000 or more ha of contiguous forest - the minimum area that studies have determined is critical for conservation of biological diversity - and both are in the Dry Zone. To address this fragmentation, preparation of management plans for clusters of wildlife reserves is being proposed under DWLC's 5-Year Investment Plan.

A NORAD and UNDP-financed conservation review to identify Wet Zone forests of high conservation value is currently being carried out by the Forest Department under the Forestry Sector Development Project, and it is expected that a significant proportion of the 8 per cent of the Wet Zone that remains under forest cover will be recommended for protected status. DWLC has recently commissioned LUPPD to compile land suitability data to support proposals to exchange wildlife and other land in several locations. A comprehensive review of protected areas under jurisdiction of DWLC that would complement the forestry review is needed.

Except for Sinharaja Forest and protected areas in the Mahaweli areas, none of the other protected areas has a conservation management plan.

The buffer zones of existing protected areas offer potential for restricted harvesting, e.g. collection of firewood and forest products, planting of forest plantations and perennial crops and controlled grazing, that would generate income for local communities and preserve adjacent protected areas.

NATIONAL ENVIRONMENTAL ACTION PLAN

RECOMMENDATIONS	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Improve the representation of Wet Zone and coastal habitats in the protected area system. This task should be accomplished by reviewing current forest, wildlife and coastal protection networks, reevaluating the biological and ecological values of existing reserves, recommending redemarcation of boundaries based on biological, ecological and social considerations, and developing conservation and management goals for each area. The ongoing forestry conservation review and activities planned under the Five-Year Investment Plan of DWLC could provide useful input to such a review. The review should consider re-allocation of lands between protection and production uses, and between institutions, and should recommend next steps toward development of reallocation procedures. This activity should be a prerequisite to preparation and implementation of conservation management programme.	Immediate	100,000	DWLC in collaboration with FD, CCD, NARA, CEA
2 Prepare and implement conservation management plans for the following seven protected area clusters, including adjacent forest reserves: - i Horton Plains Group of Reserves - ii Sinharaja Group of Reserves - iii Yala Group of Reserves - iv Wilpattu Group of Reserves - v Gal Oya Group of Reserves - vi Uda Walawe Group of Reserves - vii Knuckles Range Additional clusters could also be taken up for implementation once the current conservation review is completed and new areas of conservation value are identified. A brief analysis of experience gained under integrated protected area management programmes under the Mahaweli Environment Project should precede preparation of the plans. The plans should cover: (i) protected areas boundary survey and demarcation; (ii) habitat management and enrichment programmes;	Medium to Long-term	7 x \$ 2,000,000 = 14,000,000	DWLC/FD, + PC/LA NGOs CCD, NARA

RECOMMENDATIONS	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
[2 Cont'd...] (iii) infrastructure development and maintenance, including visitor use facilities and programmes; (iv) community-based resource management activities (buffer zones, fishing, harvesting, etc.); (v) staff training and development; and (vi) operation and maintenance. Where designated clusters include wildlife and forest reserves, DWLC and FD should be jointly responsible for plan preparation, and the plan should identify their implementation roles in detail. If indicated, the plan should provide for reallocation of protected areas. Provision should also be made for management of other forest areas that will be identified under the ongoing conservation review.			
3 Prepare a plan to develop and test buffer zone protection models, and to implement several pilot programmes in community-based resource management. Such plans should build on experience gained under the village conservation society programme initiated by a local NGO, March for Conservation, and NORAD/IUCN's Sinharaja Conservation Programme. Other test programmes could include common grazing lands around Bundala National Park (BNP) and the Floodplains National Park (FNP) where there is severe pressure for grazing. Low-level production systems for minor forest products could be tested in degraded forest reserves or sanctuaries. Local research institution could be contracted to conduct the necessary surveys and develop the plans in collaboration with the DWLC and FD.	Medium	4 x \$ 200,000 = \$ 800,000	DWLC/FD with PC/LA + NGOs + Community Groups + Universities

2

MOBILIZATION OF RESOURCES TO SUPPORT CONSERVATION

BACKGROUND:

Two potential mechanisms for generating additional resources for conservation have been successful in other countries, and should be tried in Sri Lanka: (i) a Wildlife Trust Fund for generation of extrabudgetary funds for conservation; and (ii) a programme to attract private investments in nature-based tourism.

Wildlife Trust: The Government's current period of budgetary restraint calls for innovative approaches to generating sustainable funding for conservation. Significant opportunities exist to obtain such new funds given the increasing public and international support for conservation objectives. The Government of Sri Lanka has therefore, established a charitable Wildlife Trust to: (i) develop a scheme for compensation for wildlife damage; (ii) promote public-private sector cooperation in wildlife conservation; and (iii) encourage and support conservation education and community-based resource management programmes. The Managing Board would comprise Government officials, NGOs and private sector representatives. It has its own resource allocation capability and is not responsible to the Government. A small staff would be created to administer the Trust and co-ordinate activities implemented through the Trust in collaboration with the DWLC.

Link private tourism sector to nature-based tourism: Natural areas such as mountains, forests, wetlands, coral reefs and beaches are major attractions for tourists. Tourism can bring numerous socioeconomic benefits to a country and region, by creating employment, stimulating local economies, generating foreign exchange, stimulating improvements to local transportation and infrastructure and creating recreational opportunities for local people. Properly planned and managed tourism in natural areas can be both non-polluting and renewable, and there are numerous examples worldwide where tourism has provided a powerful incentive for conserving biological resources. Inadequately managed tourism, however, involving inappropriate siting of facilities and overuse of sensitive areas, can threaten critical resources and lead to inappropriate development around protected areas.

Initiation of a nature-based tourism industry that can effectively benefit local communities and stimulate rural development requires close cooperation between several Government agencies, universities, NGOs and the private sector. Since most nature-related tourism activities depend on Government-controlled natural resources in protected areas or state forests, the Government must take the lead, directly or indirectly, in natural resource management and supporting nature-based tourism, mainly by creating a proper atmosphere for private enterprise to assume development and managerial roles consistent with sound plans for environmental conservation. This goal can be advanced if nature-based tourism initiatives are integrated into the development of Sri Lanka's overall tourist development plan, which is being initiated by the World Tourism Organization, under the auspices of the Tourist Board.

RECOMMENDATIONS	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Develop specific strategies and work plans to meet the objectives of the Wildlife Trust, including staffing needs, resources, time frame and funding needs to implement them, and develop specific funding strategies in co-operation with the public and private sector and the international donor community.	Immediate	25,000	Wildlife Trust

RECOMMENDATIONS	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
2 Commission a team of local and expatriate tourism specialists to prepare feasibility studies for nature-oriented tourism development in a few key areas where the potential for such tourism is good, involving surveys of economic potential, visitor potential and market, potential and preferred recreational activities, perspectives of tour operators, infrastructure and services needs, environmental impact monitoring and controls, and training needs of guides, tour operators and resort managers. At the same time, consult with the International Finance Corporation (IFC) to assess incentives and programmes that will attract the private sector to enter Sri Lanka's tourism sector.	Immediate	100,000	DWLC, FC, Tourist Bd., CEA, CCD, NGOs
3 Prepare and implement management plans emphasizing public-private collaboration in resource management in two test sites (terrestrial and coastal) that have available infrastructure and services, tourist potential and major attractions. The Tourist Board, DWLC, CCD, NARA, local people and private hoteliers should collaborate to develop a suitable resource zoning plan according to criteria such as infrastructure facilities, recreational and biological values of the resources, sensitivity of habitats, etc. The plan should identify: (i) interpretation facilities to be developed; (ii) service and infrastructure needs; (iii) measures to protect the resources; (iv) roles and responsibilities of the different partners; and (v) a package of incentives to the entrepreneur to run a nature-based tourism facility, including means for the neighboring people to be provided employment opportunities in the enterprise, and for returns on investment to be ploughed back to resource conservation efforts in the area. Suggested areas include the Victoria-Randenigala-Rantambe Sanctuary, the coastal Bundala Sanctuary or Pallemalla Sanctuary, and Hikkaduwa Sanctuary, which is currently under threat from overuse.	Medium	200,000	DWLC, CCD, NARA, PC/LA, Hoteliers, Communities, NGOs

3

PLAN FOR THE CONSERVATION AND MANAGEMENT OF THE ELEPHANT

BACKGROUND:

The long term-survival of the Sri Lanka elephant population—estimated at about 3,000—has been an issue of increasing concern to the Government and the people. The opening of new lands for agriculture has increased human/elephant conflict in most parts of the Dry Zone of Sri Lanka. Effective programmes should therefore be designed and implemented to ensure the conservation of the elephant in perpetuity. Three major programmes can be identified for the conservation of the elephant:

- * Managing viable elephant populations in suitable habitats (on-site management);
- * Capturing and managing elephants in captivity or translocating to suitable alternative habitats (capture management); and
- * Promoting public awareness and soliciting community support for elephant conservation (outreach).

In order to ensure success in the long-term conservation of elephants all three programmes should be effectively integrated and implemented in conjunction with each other.

Under the proposed 5-Year Investment Plan of the DWLC, an experimental programme for the management of the human/elephant conflict will be implemented for the conservation of the elephant. This programme will work out a combination of strategies and test their effectiveness in specific field situations. The plan of activities include: (i) development of standardized methodology for crop damage assessment; (ii) development of approaches to manage human/elephant conflict; (iii) enhancement of opportunities for cooperative action between public agencies, the private sector and communities; and (iv) testing in-situ and ex-situ approaches.

RECOMMENDATIONS	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Plan and implement a comprehensive elephant management programme for Sri Lanka emphasizing on-site management, captive management and outreach. The components of the programme would include: <u>A Captive Management Programme</u> i Elephant Conservation and Management Center for elephant maintenance, breeding, training, veterinary care and research and related infrastructure including staff housing, office and storage space;	Medium to Long-term	3,000,000	DWLC with FD, PC/LA, ED

RECOMMENDATIONS	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
[A Cont'd...] - ii Satellite Centers for maintaining elephants near existing protected areas (6 satellite centers) with facilities for elephant training and veterinary care and related infrastructure; - iii Personnel training and public education; - iv Equipment and Supplies including vehicles, laboratory equipment, field equipment, etc.. B <u>On-site Management</u> - i Elephant control and management units (12) and related infrastructure such as staff housing, office and storage space; - ii Personnel training and public education; - iii Equipment and Supplies including vehicles, field equipment, etc.; - iv Habitat Management and manipulation, electric fencing, etc. C <u>Outreach</u> - i Programme Development for public education, crop protection, community participatory programmes; - ii Personnel Training; - iii Visitor Centers, auditoriums and related staff housing, recreation facilities, etc.; - iv Equipment and Supplies including vehicles, audio-visual equipment and supplies, interpretation facilities and equipment and library facilities.			

WETLAND CONSERVATION**BACKGROUND:**

Sri Lanka possesses three hectares of inland lentic water for every square km of land [de Silva, 1988]. It has one of the highest densities of inland lakes, tanks, ponds, marshes and other still water available in any country in the world. Not surprisingly, with its ancient hydraulic civilization and emphasis on self-sufficiency in rice production, about 95 per cent of Sri Lanka's inland freshwater habitats are man-made, comprising mainly rice paddies and village reservoirs. The functions and values of aquatic resources, particularly wetlands and floodplains, have received increasing international attention over the past two decades. Recognizing the importance of its wetlands, Sri Lanka is a signatory of the international treaty convention for wetlands conservation (Convention of Wetlands of International Importance, Ramsar, 1971).

Wetlands require protection for various reasons. They store floodwater and release it slowly, thereby lowering flood peaks and allowing downstream areas to be protected. In urban areas subject to rapid storm water runoff and flooding, wetlands, particularly in the southwestern coastal zone, serve this important function as "sponges" by absorbing floodwaters and mitigating downstream property damage. Although not yet measured in Sri Lanka, a U.S. study has estimated that loss of urban wetlands around the Charles River (near Boston, Massachusetts) would cause flood damage averaging about U.S. \$ 17 million per annum [State of the Environment, Conservation Foundation, 1987]. Other important functions of wetlands include fishery production, wildlife habitat (including habitat that supports major bird as well as elephant populations), livestock and grazing, and water for irrigation and other uses. Urban wetlands, in particular, also provide recreational open space and attractive habitats for birds and wildlife.

Many rural communities around wetlands depend on them to meet their basic needs. Actions are needed to help such communities develop solutions to their economic problems. By not recognizing and conserving wetland values, communities located around coastal lagoons in the southern coast, such as Rekawa, Kalametiya, and Lunawa, suffered setbacks from costly technical mistakes which resulted in a drastic reduction of fisheries in the lagoons, changed salinity and hydrological regimes, and the consequent reduction in the economic potential of coastal wetlands.

An important result of the National Conservation Strategy -Action Plan is the establishment of a National Wetland Steering Committee (NWSC) by the CEA, with participation of agencies involved in conservation and development agencies and NGOs. The necessary legal framework for comprehensive wetland protection does not yet exist, however. To fill this gap a number of actions are required that the NWSC can encourage once it is fully established.

RECOMMENDATIONS	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Compile a national inventory of wetland resources which would, inter alia, provide information on the nature, extent, current protection status, and potential importance of different wetland areas in the island. This should supplement the work already done on Sri Lanka and listed under the Asian Wetland	Immediate	50,000	NWSC and Universities, NGOs

RECOMMENDATIONS	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
[1 Cont'd...] Inventory. The updated inventory should provide the basis for determining priorities in wetland management and evaluate the impact of major development programmes on wetland ecosystems. The NWSC should coordinate this effort, which could be commissioned to an NGO or a university, and ensure linkage with the Dutch-supported Wetlands Conservation Study.			
2 Prepare a National Wetland Master Plan which would incorporate the coastal marine and inland freshwater wetlands. The Master Plan should form the basis on which wetland issues are addressed. It should identify policies, goals, institutional arrangements, land and water use guidelines, and necessary legislation to ensure effective wetland practices and protection. The Plan should explicitly provide opportunity and incentives to local residents to participate actively in the management of inland aquatic resources, in particular. The Wetland Steering Committee should coordinate the preparation of the Master Plan with inputs from the following agencies: Coast Conservation Department, CEA, National Aquatic Resources Agency, Natural Resources, Energy, and Science Authority (NARESA), Department of Wildlife Conservation, and the Land Use Policy Planning Division (LUPPD).	Medium	200,000	NWSC with CCD, CEA, NARA, NARESA, DWLC, LUPPD NGOs Universities
3 Develop private-public partnerships for sustainable management of critical wetland sites. To develop such private-public partnerships to manage wetlands effectively, pilot Special Area Management (or SAM) plans for key coastal lagoons in the south should be formulated and carried out.	Medium	200,000	NWSC with CCD, NARA, DWLC, LUPPD, Tourist Bd., PC/LA, NGOs, Local Communities, Private Sector

RECOMMENDATIONS	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
[3 Cont'd..] Projects in the south are appropriate because, unlike lagoons on the western coast, degradation has not proceeded beyond repair, and the southern region is the focal point of a great deal of new Government activity.			
4 To design and plan SAMs, the following steps must be taken initially: * Field surveys to evaluate resource values, major threats, and conservation significance, feasibility and needs; * Socioeconomic surveys including land use and land ownership; * Field surveys to identify habitat diversity, abundance or resources such as commercial fishes, and fish species of endangered, rare, or depleted status; and * Surveys to determine the benefits from and beneficiaries of water sports and other recreational activities, tourism development, and the impacts of such activities.	Medium	50,000	NWSC with Universities, NGOs
5 Identification, formulation, and implementation of coastal Special Area Management (SAM) plans should be the responsibility of the Coastal Conservation Department and the Department of Wildlife Conservation, depending on the location and status of the particular area. Agencies contributing to these efforts would be NARA, CEA, Tourist Board, Forest Department, LUPPD, Local and Provincial Councils, Pradeshiya Sabhas, and NGOs. Such pilot efforts with appropriate technical assistance and local training programmes would generate information which could then be replicated elsewhere.	Medium	250,000x4	NWSC with NARA, CEA, Tourist Bd., FD, LUPPD, NGOs, PC/LA

URBAN POLLUTION

OVERVIEW

Twenty-one percent of Sri Lankans live in urban areas¹, which cover a mere 0.5 per cent of the total land area of Sri Lanka. Only one Sri Lankan city, Colombo, has a population of more than 500,000, and nine other urban areas exceed 50,000 population. Aggregate urban growth rates have been low--1.2 percent (less than the population growth rate) in the 1970s, and 2.4 percent in the 1980s.

The environmental impacts of urban and industrial growth in Sri Lanka derive mainly from the concentration of growth in the Colombo area. While the aggregate urban growth rate hovered around 1.2 percent in the 1970s, localities in the Colombo Urban Area recorded 10-40 percent growth in the same period. This could be attributed to the fact that the Colombo City is already getting saturated as all the buildable land is extensively built up and only a marginal increase of population can be accommodated without major re-development and reclamation programmes. Therefore, the shortage of land and high prices within the Colombo City compel people to move into the suburban areas, where land is available comparatively at reasonable prices. This urban sprawl could result in urban problems, long commuting distances and densification of semi-urban areas without supporting infrastructure. Eighty percent of manufacturing enterprises and over 60 per cent of registered vehicles are located in Colombo. Concentrated population and industrial growth have overwhelmed the capacity of municipal institutions and stream protection regulations to protect the rivers, canals, low-lying lands and marshes in the Colombo area, which are essential for regulating surface and groundwater supplies, controlling flooding and maintaining soil bearing capacity.

UDA is the principle agency responsible for the planning, development and control of urban activities within urban areas.

The principal environmental issues in the urban sector are as follows:

- (1) Inadequate domestic and solid waste disposal;
- (2) Loss of urban stormwater basins;
- (3) Inadequacy of planning tools and institutional framework for addressing the above.

¹Only four Asian countries--Vietnam, Thailand, Bangladesh and Nepal--are less urbanized than Sri Lanka.

1

INADEQUATE DOMESTIC AND SOLID WASTE DISPOSAL

BACKGROUND:

Colombo is the only city in Sri Lanka with a sewerage system, which is connected to about 60 percent of residences in the jurisdiction of the Colombo Municipal Council, and about 25 percent of residences in the Colombo Urban Area, in 1986. Twenty-two percent of households use septic tanks, and 10-15 percent of households (about 65,000 - 90,000 inhabitants) have no toilet facilities and discharge sewage directly into the surface water network. About 50 percent of Colombo's population is in the low-income category and lives in semi-permanent dwellings, where high spot densities strain existing waste disposal infrastructure and make placement of septic tanks relative to drinking water sources difficult to control. About 150,000 residents of Colombo District were at potentially high risk of contracting disease from contaminated water in 1981; by 1991, that number was estimated to have increased by about 12 percent. Until recently, sewage flow into the Kelani River from the city's northern district ranged from 70,000 to 90,000 m³ per day, and this constituted at least 50-60 percent of the total organic load discharged into Colombo's canal network. Centrally collected sewage from Colombo is discharged untreated to the sea through the recently constructed Colombo north and Colombo south ocean outfalls 1.5 km from the city center, but the outfalls may only have transferred the pollution problem to a marine and coastal environment, and increased volumes of sewage raise the possibility of pollution of the adjacent coastline.

In cities and suburban areas outside Colombo, such as Kandy, on-site sewage disposal systems serve only 50-60 percent of households, and the remainder, which are concentrated in high density settlements at the city core, discharge sewage directly into surface waters. In Jaffna, Kandy and parts of Moratuwa, faecal coliform contamination of groundwater is evident, and alternative water supply sources need to be considered. NWSDB and Urban Local Authorities are exploring the possibility of sewerage schemes for some major cities, and local authorities' lack of institutional capacity to plan and manage sewerage disposal systems represents a major constraint.

About 1200 tons of solid waste were generated in the Colombo urban area daily in 1990. Per capita generation of solid waste in the Colombo municipality increased from 0.75 kg/person/day to 0.98 kg/person/day (30 percent) from 1986 to 1990, and is expected to increase by another 21 percent by 2010. About 90 percent of solid waste in Colombo Municipality is collected or brought by households to temporary collection sites, but there the garbage typically remains unattended for considerable periods, creating a health hazard and nuisance to residents. Some smaller Urban Local Authorities collect as little as 5 percent of solid waste. Low-lying lands are used as disposal sites, and much of the waste finds its way into drains and watercourses, where it causes deterioration of water quality and exacerbates flooding.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Conduct a cost-benefit analysis of Colombo's two sewage disposal options: construction of a pre-treatment plant and/or lengthening the ocean outfalls.	Immediate	100,000	CMC/ NWS&DB
2 Prepare a strategic plan for solid waste management in the Colombo urban area, including technical, planning and administrative (inter-governmental co-ordination) aspects.	Immediate	100,000	CMC/UPU
3 Conduct a study to determine existing water pollution levels in selected urban centers outside Colombo, such as Kandy and Jaffna. The study should: (i) design and initiate regular water quality monitoring in these urban areas; and (ii) assist in establishing priorities for sewerage schemes in cities outside Colombo.	Immediate	150,000	CEA
4 Identify: (i) priority sewerage and solid waste development needs in faster-growing urban centers outside Colombo, based on water quality and health data; (ii) costs of recommended sewerage and solid waste management investments; and (iii) an institutional strategy for developing sewerage and solid waste management units in urban centers outside Colombo.	Medium	Substantial Investments	UDA

2

LOSS OF LOW-LYING URBAN WETLANDS AND STORMWATER BASINS**BACKGROUND:**

In Colombo and other coastal cities, particularly along the urban-industrial corridor that extends along the southwest coast of the island, urban population growth and skyrocketing land prices are exerting intense pressure on the network of coastal marshes on which Colombo and other coastal cities depend for drainage and protection from flooding. Filling of over 200 hectares of marshland in these urban stormwater basins since 1968 has exacerbated flooding in low-lying areas of the city. Many of the 50 percent of Colombo residents with low incomes occupy marginal land, including canal and stream reservations that are legally protected and whose protection from obstructions is critical to drainage and flood protection. A highly visible symptom of the disruption of drainage in Colombo that has resulted from filling of marshes and occupation of stream and canal reservations is the eutrophication of Beira Lake, which receives waste from several outfalls with COD levels up to 800 mg/l and suspended solids up to 300 mg/l.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Design and implement a program for rehabilitation of Colombo's natural drainage system, focusing on restoration of Beira Lake and including designation of beneficial uses, a strategy to address marsh filling and stream and canal reservation settlement, and identification of potential investments for tourism and recreational use.	Immediate	1,500,000	MEIP/CEA
2 Prepare and implement a demonstration project to promote environmental improvements in low income settlements along selected canal banks in Colombo City.	Immediate	20,000/ settlement	MEIP
3 Prepare restoration plans for urban water bodies subject to deteriorating water quality in cities other than Colombo, such as Kandy Lake, including proposed institutional mechanisms for implementing the plans and addressing extra-urban pollution sources, such as agrochemical runoff.	Medium	500,000	Kandy MC

3

LACK OF INSTITUTIONAL CAPACITY TO INCORPORATE ENVIRONMENTAL PARAMETERS INTO URBAN PLANNING AND O+M

BACKGROUND:

The Urban Development Authority (UDA) is the principal agency responsible for urban planning and development. Urban master plans produced by UDA, such as the Development Plan for the City of Colombo, tend to emphasize infrastructure to support urban growth, and to give insufficient attention to limits to growth, such as urban and industrial waste disposal capacity, and elements of the urban natural resource base that require protection to ensure sustainable urban growth, such as stormwater basins. Urban development plans also typically fail to calculate the costs of protection and maintenance of urban infrastructure and to provide mechanisms for generating resources for protection and maintenance. The issue here is one of familiarizing urban planners and governments with key urban environmental planning issues and best practice, and developing institutional capacity to implement best practice.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Prepare an institutional strategy for development of an urban environmental planning and O+M capability in UDA and other relevant agencies in Colombo and selected other urban areas. The strategy should give priority to urban waste treatment and disposal, protection of urban drainage networks, development of urban pollution and environmental protection cost recovery mechanisms, and integration with industrial siting and other industrial pollution control measures.	Immediate	300,000	UDA
2 Implement the urban environmental institutional strategy, including establishment and preparation of a work plan for an environmental cell in UDA, a study of urban environmental cost recovery options, etc.	Medium	150,000	UDA

INDUSTRIAL POLLUTION

OVERVIEW

Manufacturing industry accounts for less than 20 percent of Sri Lanka's GDP. Manufacturing industry in Sri Lanka is comprised of: (a) a few large-scale heavy industries, including petroleum, cement and paper, most of which are publicly owned, use outdated technologies (dating from the 1960s or before), and are located away from urban areas; and (b) small and medium industries, which are largely privately owned and newer than the public sector enterprises. The liberalization of the economy in 1977 led to 6 percent per annum industrial sector growth during the period 1980-87, including significant growth in the textiles, food-processing and leather subsectors. The Government's overall development strategy looks to rapid expansion of the manufacturing sector as the main engine of economic growth. The New Industrialization Policy (NIP) of 1989 calls for deregulation and devolution of public sector enterprises to foster a more competitive atmosphere, and provision of infrastructure, management improvement assistance and incentives to reduce production costs. This NIP has been complemented by the issuance of the Industrial Promotion Act No. 40 (1990).

Eighty percent of small and medium industries are concentrated in Colombo and neighboring Gampaha district, and their main impacts are on water quality. Estimation of the severity of industrial pollution in Colombo and Gampaha districts is difficult, due to poor water quality data. What limited data are available indicate that the Kelani River, which supplies most of Colombo's potable water and receives a significant proportion of domestic and industrial wastes from Colombo and Gampaha districts, had dissolved oxygen levels of 18-28 percent saturation in the last several kilometers before its discharge to the sea as long ago as 1973-74, and that an estimated 25 percent of the organic waste load comes from industry (versus 75 percent from sewage).^{2/}

There are also localized industrial impacts on air quality, most visibly from Sri Lanka's three cement factories, which are outside large urban areas (Puttalam, Kankesanthurai, Galle) and on soils and water quality due to mineral extraction. The focus here, however, is on water pollution due to untreated or inadequately treated industrial effluents in Colombo and Gampaha districts.

^{2/} With the opening of two marine outfalls that channel Colombo's wastes to the sea and bypass the Kelani, the proportions of organic load attributable to industry and sewage may have changed.

The Government's main policy responses to industrial pollution include the following:

- (a) an Environmental Protection Licensing Scheme (1990) covering existing industries. Under the scheme, existing industries must obtain a license from the CEA, which includes conditions requiring the industries to control their effluents to an agreed level within a specified time period. Some 361 industries received licenses during the first year of implementation of this scheme. Implementation of the licensing scheme is constrained, however, by lack of trained staff, inadequate standards, lack of incentives, unreliable data, and lack of an enforcement strategy; and
- (b) mandatory Environmental Impact Assessments (EIAs) of all proposed industrial ventures involving potential pollution. National (Environmental Impact Assessment) Regulations have been framed and have undergone public review, and are expected to become law in July 1991. Under the Regulations, an Initial Environmental Examination (IEE) or EIA report will have to be submitted to the relevant project approving agencies as a condition of project clearance. The same constraints as above--lack of trained staff, inadequate standards and guidelines, lack of incentives and lack of reliable data--are present here.

The principal environmental issues in the industrial sector are as follows:

- (1) gaps in support for main regulatory policies:
 - * tools for environmentally-sensitive siting are poorly developed;
 - * incentives to complement pollution control regulations and accomplish environmental protection objectives more cost-effectively;
 - * technical expertise and treatment facilities;
 - * water quality monitoring data; and
- (2) lack of data on environmental and health impacts of industrial pollution. Without such data it is difficult to assign appropriate priorities to pollution control issues and make pollution control strategies cost-effective.

1

TOOLS FOR ENVIRONMENTALLY-SENSITIVE INDUSTRIAL SITING

BACKGROUND:

Industrial siting in the urban areas of Sri Lanka is based upon the UDA zoning plan which includes various landuse categories such as: primary residential, mixed residential, commercial, general industrial, special industrial, open space, deferred, and others. For each landuse category, detailed guidelines regarding proposed development actions are available, especially as related to planning and building regulations. As for general and special industrial zones, limitations are placed according to the type of industrial activities. Size limits have been placed on cottage industries located in residential zones. Only the City of Colombo has a legally approved development plan; others are still in the process of being developed.

While UDA planning controls and regulations are generally well-defined, their implementation has proved to be difficult. An important source of that problem is the paucity of clear environmental criteria for industrial siting and industrial estate planning in these regulations. To overcome this problem, the NBRO prepared its Environmental Impact Assessment Manual in conjunction with UDA planners. The Manual classifies industries according to pollution levels, introduces a step-by-step procedure for evaluating different classes of industries, and provides guidelines for siting of these different classes.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Commission a study to assess the environmental suitability of sites identified under the Industrial Promotion Act (1990) and rank them in order of suitability.	Immediate	100,000	M/IS&T
2 Develop EIA guidelines and criteria for industrial development projects including estates. Such guidelines should supplement the industrial estate planning methodology prepared by NBRO.	Immediate	NAREPP	CEA
3 Ensure that NBRO's Environmental Assessment Manual is used by project proponents and UDA in preparation and review of EIAs. Add to the NBRO Manual a description of credit institutions' role in environmental assessment of small- and medium-scale industrial projects.	Immediate		UDA

2

INDUSTRIAL POLLUTION MANAGEMENT**BACKGROUND:**

The major tasks of the licensing scheme are the responsibility of the CEA, but powers have been devolved to the Greater Colombo Economic Commission (GCEC) for licensing industrial establishments in their area of authority. Licensing procedures are well defined, and implementation began in July 1990. However, there are constraints on implementation which include inadequate standards, monitoring systems and testing facilities.

The present strategy of pollution control through regulation alone may prove burdensome and not cost-effective. Financing of pollution abatement needs to be planned. The majority of industries in Sri Lanka are small- to medium-scale operations. Many are reluctant or unable to make any investment outlays for pollution control. An approach combining best practicable criteria and time targets with some form of financial assistance and/or incentive may be the most reasonable one for these industries. For large scale industry with higher investment levels that can better afford the requisite pollution control technology, a different strategy incorporating effluent fees may be appropriate.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Review and update existing discharge standards and pollution control guidelines for all sectors of industry, incorporating a best achievable practice approach. The standards developed should have a time horizon with progressive stringency on discharge limits.	Immediate	40,000	CEA
2 Study and propose mechanisms--linked with CEA enforcement--for financing pollution abatement through subsidies, incentives, loans, and/or grants. The proposed mechanisms should include eligibility criteria, period of operation, etc. If indicated, identify next steps toward establishment of an industrial pollution control fund.	Immediate	100,000	M/PPI

3

WASTE ABATEMENT EXPERTISE AND FACILITIES

BACKGROUND:

The majority of industries in Sri Lanka discharge wastes without treatment. Among those that are equipped with waste treatment equipment, lack of technical expertise in waste abatement has led to adoption of ineffective treatment technologies. Existing waste treatment plants are in some cases overdesigned, and are not operated efficiently due to lack of knowledge. Regular monitoring of intake and effluent and record-keeping are lacking. The involvement of the private sector in providing consulting and advisory services is limited as industrial pollution control is still not considered to be a profitable business activity. The Dutch Government has financed a feasibility study for a demonstration rubber waste treatment plant.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Conduct an assessment of the requirements for pollution control technologies for the worst polluting industries in Sri Lanka. The rationale for the identification of the worst industries should be analyzed and explained. The study should also identify the amount of in-house and external expertise and facility needed.	Immediate	150,000	CEA
2 Strengthen the research capabilities of Ceylon Institute of Scientific and Industrial Research (CISIR) on appropriate waste treatment technologies and process improvements and its capacity to transfer this know-how to the private sector, GCEC, the CEA and other agencies.	Immediate	100,000	CISIR
3 Provide designs of pre-treatment facilities for at least 20 major polluting industries in the Colombo metropolis.	Immediate	100,000	MEIP
4 Study and implement the feasibility of constructing a pipeline from Biyagama to Kelani River.	Immediate	150,000	GCEC

4

WATER QUALITY MONITORING AND RELATED ISSUES

BACKGROUND:

Present water quality monitoring practices are poor, and consist of scattered, irregular sampling and analyses, which neither cover all the principal water quality parameters that should be monitored in a given water body nor provide time-series data. Regular monitoring is essential to development of a cost-effective industrial pollution control strategy and effective enforcement of emission standards and best practice guidelines. To ensure that results from the monitoring are linked to standard-setting and enforcement efforts, design and implementation of a monitoring system should proceed in stages.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Identify sampling points in key areas where industrial development is planned, and collect, analyze and publish quarterly water quality data, showing comparisons with any applicable Sri Lankan and other country standards.	Immediate	25,000	CEA
2 Prepare and implement a workplan for development of a water quality management model for the Kelani River based on designated uses, including estimates of required staff resources, legal framework (regulations and incentives) and an analysis of the existing capacity of the institutions designated to carry out the monitoring.	Beyond 1996	1,000,000	M/LI&MD
3 Expand designation of beneficial uses and the water quality monitoring network to other priority water bodies in Sri Lanka.	Beyond 1996	N/A	M/LI&MD

ENERGY

OVERVIEW

Principal primary energy sources in Sri Lanka include: (a) biomass (71 per cent); (b) hydroelectricity (10 per cent); and (c) petroleum (19 per cent). Biomass, the dominant fuel, is untraded, except for urban cooking fuel and woodfuel consumed by industry. High rainfall and the hilly topography of the southwest provide Sri Lanka with excellent hydropower potential, and hydro capacity expanded nearly threefold in the period 1977 to 1988. All oil is imported. Rising oil prices caused severe strains on the Sri Lanka economy in the 1970s and '80s, leading to acceleration of hydro development and a shift by households and small industries from oil to woodfuel and LPG.

The principal environmental issues in the energy sector are as follows:

- (1) Biomass, which includes fuelwood and agricultural residues and is mainly used for household cooking and industry, constitutes 71 per cent of primary energy, but contributes only about 40 per cent of usable energy. Biomass fuel consumption is growing at about 1.4 per cent per year, i.e., at approximately the overall population growth rate, and biomass is expected to remain the dominant fuel in the medium term. To relieve additional pressure on forests, biomass fuel conservation measures are essential;
- (2) Consumption of electricity grew by 5.8 per cent annually from 1980 to 1990, and predicted annual power system growth from 1991 to the year 2000, which is expected to be driven by industrial growth, is 7.3 per cent. There is no regular monitoring of emissions from Sri Lanka's three existing oil-fired thermal plants, nor has Sri Lanka enacted emission standards for thermal plants. To promote an environmentally beneficial mix of power sources and improved commercial energy efficiency, integrated energy planning and industrial energy conservation measures are needed. In light of plans to invest US\$1 billion in the 1990s toward development of a mixed commercial power system with roughly equal hydro and thermal capacities by the year 2007, GOSL capacity to assess, monitor and mitigate potential adverse environmental impacts of hydro and thermal power plants needs strengthening;
- (3) Inefficient combustion of petroleum in motor vehicles is a primary cause of growing air pollution in the Colombo area. To prevent adverse health and economic impacts, initial steps to develop incentives and regulations to reduce air emissions need to be taken.

A more detailed presentation of these issues follows.

1

DOMESTIC FUELWOOD CONSUMPTION**BACKGROUND:**

Biomass, including agricultural residues and fuelwood, accounts for 71 per cent of primary energy consumption in Sri Lanka, and dependence on biomass fuels is pervasive: 90 per cent of Sri Lankan households use biomass for cooking, and biomass constituted slightly less than 70 per cent of industrial fuel in 1989. Biomass fuel consumption becomes an environmental issue when it involves unsustainable cutting of natural forest.

Households consume 87.6 per cent of biomass consumed in Sri Lanka, and an estimated 20-25 per cent of biomass consumed by households is fuelwood from natural forest. Household fuelwood demand is estimated to be increasing by 1.4 per cent per year, i.e., at about the population growth rate. There are already local fuelwood shortages in the hill country and Colombo, and a national fuelwood shortage is predicted by the year 2000. Although biomass accounts for 71 per cent of primary energy consumption, it provides only 40 per cent of useful energy. The principal reason for this is the low thermal efficiency—less than 15 per cent—of the traditional 3-stone open hearths that are used by nearly 90 per cent of households. The principal effort to address unsustainable household fuelwood consumption has involved dissemination of efficient stoves. More efficient stoves can improve stove efficiency up to 20 per cent, and reduce average household fuel consumption from 2.3 MT/year to 1.7 MT/year. Stoves developed and disseminated under the National Fuelwood Conservation Programme (NFCP) have been installed in 300,000 households since 1984. However, despite a significant subsidy of Rs.45 per household (75 per cent of stove costs), voluntary household participation in some areas is as low as 25 per cent, and 10 per cent of households discard the stoves within 2 years because of their inconvenience.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Extend the stove dissemination programme by about 250,000 for 1991-93 while, at the same time, initiating an evaluation of past experience which would include the following: * Review experience under the National Fuelwood Conservation Programme (NFCP); * Evaluate constraints to stove dissemination by region, and propose ways by which demand can be stimulated; * Estimate costs and benefits of an expanded dissemination programme by region, as compared to other options such as expanded farmer woodlots; and * Propose a cross-sectoral (forestry and energy) mechanism under the Ministry of Environment and Parliamentary Affairs (MEPA) to monitor fuelwood supply and demand, and the impact of conservation measures. If the evaluation results are favorable, a proposal to expand the stove dissemination programme beyond the present plans should be prepared.	Immediate	300,000 for dissemination 40,000 for evaluation	M/EC, CEB Investigate greater NGO participation

BACKGROUND:

Industry uses only one-sixth the biomass fuel consumed by households in Sri Lanka. While estimated household fuelwood extraction from natural forests amounts to 21 per cent of annual biomass fuel consumption in Sri Lanka, estimated industrial fuelwood extraction from rubber plantations and natural forests is 4-5 per cent of annual biomass consumption. However, industry's dependence on biomass, which has been about 68 per cent, is high relative to other countries. The largest industrial biomass fuel consumers in Sri Lanka are the tea industry (33 per cent), hotels/restaurants (15 per cent), brick/tile makers (13 per cent), coconut and tobacco processors (11 and 10 per cent respectively), bakeries (8 per cent) and rubber processors (6 per cent). Availability of agro-residues for industrial fuel is limited to 1 million MT/year. Forty-nine per cent of biomass fuel consumed by industry is rubber wood and 38 per cent is from natural forests. Low fuelwood prices relative to other energy alternatives have encouraged conversion of oil-fired boilers and furnaces to fuelwood use.

Measures to improve industrial fuelwood efficiency and expand fuelwood supplies have included: (a) state- and donor-funded energy audits of tea and brick/tile establishments; and (b) an initiative to establish several thousand hectares of dedicated fuelwood plantations by several tea factories.

Industry consumed 36 per cent of electricity sold by CEB in 1989. Measures to improve industrial energy efficiency have included: (a) a National Energy Demand Management and Conservation Programme (NEDMCP) and other programmes under Ministry of Power and Energy; (b) training of energy auditors; (c) industrial energy conservation publicity and plant-level sensitization; (d) an annual Rs. 5 million Energy Conservation Fund; and (e) establishment of a small Energy Conservation Project Ministry.

Although training under the NEDMCP and other programmes developed in-house energy auditing capacity in many energy-consuming institutions, follow-up of the audits of more than 40 large and medium industrial and commercial establishments between 1984 to 1990 has been poor. A 1989 survey of audited industries showed that less than 10 per cent of predicted savings have been realized under the few conservation programmes that have been implemented. Establishment of industrial fuelwood plantations has proceeded very slowly, and there are no Government guidelines or incentives regarding industrial use of fuelwood.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Conduct a study to: (a) review industrial energy conservation experience under the National Energy Demand Management and Conservation Programme (NEDMCP) and other programmes; (b) assess constraints on and results of implementation of audit recommendations; and (c) recommend financial packages to implement identified conservation measures.	Immediate	250,000	M/P&E
2 A study to analyze price and other incentives for oil- to- fuelwood substitution, and where fuelwood consumers are getting their fuelwood supplies.	Immediate	250,000	M/P&E
3 Assist large fuelwood and electricity users to implement audit recommendations and target groups of small and medium-scale industries. Link to land allocation initiatives under EAP.	Medium	250,000	M/P&E FD

3

CONSERVATION: INSTITUTIONAL CAPACITY**BACKGROUND:**

The Project Ministry of Energy Conservation (MEC), which was established in 1990, has only four staff in addition to the Minister and severely limited resources. It is not a statutory body and its medium to longer-term role is not yet clear. A more permanent energy conservation unit, which could either form the core of the Project Ministry or be located elsewhere within the Ministry of Power and Energy, needs to be established with adequate resources.

The Government is currently considering establishment of an Energy Advisory Service that would mainly cover household energy use. Meanwhile, there is no locus of responsibility within the Government for follow-up to completed audits of large and medium industries, and no assistance for identification of energy conservation projects or development of energy management capabilities of small and medium industries. Although there is significant in-house energy auditing and management capability within many large energy consuming entities, it is not being utilized. There is only one national NGO interested in Energy Conservation which is working in close coordination with Government conservation efforts.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Develop an Energy Advisory Service to assist industries (and the household sector) to: (a) review past conservation experience under the Energy Demand Management and Conservation Programme; (b) provide technical advice to industries and commercial facilities, including both biomass and electricity consumers, initially emphasizing the small and medium-sized industries; and (c) help develop industrial energy conservation financing packages/programme.	Immediate	125,000 (initial) 75,000/ year (recur-rent)	M/P&E

4

POWER SYSTEM EXPANSION AND EIA/ENVIRONMENTAL MANAGEMENT CAPABILITY

BACKGROUND:

Sri Lanka now has cascaded hydropower reservoirs across 3 river valleys with total 1115 MW capacity, and 3 thermal plants, providing back-up capacity of 250 MW. With electricity demand expected to grow 7-8 per cent annually through the year 2000, generation and transmission systems have to expand, and planned 1991-2000 investments on new generating plants total US\$1 billion. Given that all major hydro sites have already been or are being developed, development will be toward a mixed hydro-thermal system, with equal hydro and thermal shares by about the year 2007 and the thermal share increasing thereafter. Two environmental issues related to power system expansion are: (a) assessment and mitigation of potential environmental damage from present and future thermal and hydro plants and transmission system expansion; and (b) catchment management of reservoirs, to reduce siltation of reservoirs and prolong their useful life. Although environmental impact assessments (EIAs) have been performed for some power system expansion projects in the last several years, CEB depends entirely on hired expertise to plan and integrate these studies into project design, and lacks capacity to follow recommendations of the studies. Neither catchment management nor effective liaison by CEB with other agencies effectively responsible for catchment is now in place.

CEA is currently preparing legislation about EIA studies to cover all projects which will stipulate mandatory requirements. Power generation projects will also be covered by this legislation and will form the basic guideline to the process of environmental assessment.

It is therefore, required for CEB to urgently establish an Environmental Cell and staff capability.

An energy EIA issue that should be one of the first tasks to be addressed by the CEB Environmental Cell is the incorporation of EIA preparation into overall energy project preparation. EIAs are being prepared for some, but not all, current power generation and distribution projects - the Sapugaskanda diesel-fired generating plant extension, for example, is apparently not undergoing environmental assessment. Means of ensuring that EIA results are incorporated into project design in a timely manner are uncertain. As is well known, the proposed coal-fired plant at Trincomalee aroused environmental controversy in the mid-1980's. The EIA was determined not to have adequately addressed the issues, and the environmental clearance was consequently withheld. The subsequent controversy around the proposed coal-fired plant at Mawella in 1990 illustrated the continuing lack of, and need for, an agreed EIA process for energy projects, which would allow for satisfactory public participation and be adequately linked to project design and implementation. The CEA is currently preparing legislation about EIA studies to cover all projects which will stipulate mandatory requirements. Power generation projects will also be covered by this legislation and will form the basic guideline to the process of environmental impact assessment.

NATIONAL ENVIRONMENTAL ACTION PLAN

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Set up an "Environmental Cell" in Ceylon Electricity Board, with responsibilities for: (a) development of an organizational structure within CEB and a work programme, which would include training of in-house staff in environmental audits of generating (thermal and hydro) and transmission systems; and (b) a study to estimate the present siltation rates, the total accumulated silt in power reservoirs, and how siltation affects the operation of the reservoirs. Once this cell is established, it should --with CEA-- develop a system in which environmental assessment of new projects will be undertaken. Also, it will be important to specify interactions between CEB and other Government and non-Governmental agencies for EIA studies, implementation and operation of generation, and other facilities.	Immediate	195,000 (initial) 25,000/ year (recur- rent)	CEB
2 Develop staff capability to undertake detailed assessment of environmental impacts of new projects, and co-ordination of work of subject specialists who are -- from time to time -- involved in project feasibility studies. Also introduce improved methods of incorporating environmental costs in long-term system development and operations planning.	Medium		CEB

5

INTEGRATED NATIONAL ENERGY - ENVIRONMENT PLANNING**BACKGROUND:**

An Integrated National Energy Planning capability was established in the Ministry of Power and Energy in 1984. It consisted of computer models that analyzed energy-economy interactions, incorporated utility plans in the national energy plan, and developed an integrated national energy plan for several years ahead, and staff capability to run the models, analyze policy options and make recommendations. Policy makers have made little use of this analytical tool, however, except for a few academic exercises. Without this analytical tool, therefore, each energy sector utility (electricity, petroleum and gas) prepares its own forecast for demand and its own plans to develop facilities, there is no coordination with Department of Forestry's fuelwood and other biomass development plans, and important decisions on issues such as energy substitution and pricing policy are implemented without fully understanding their likely social, economic and environmental impacts.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Re-establish the Integrated National Energy Planning capability within the Ministry of Power and Energy, with the Government's commitment to use the tools available within the Ministry of Power and Energy for policy analysis before major policy decisions are made. Ten professionals who received overseas training and 30 professionals who received in-country training in energy planning are available to implement national energy planning. No new institutional structure is required, but the working group should be made up of permanent staff.	Immediate	50,000	M/P&E
2 Extend the national energy planning function to include the evaluation of energy-environment interactions and to advise the decision makers of the implications of policy decisions.	Medium	25,000	M/P&E

6

ENERGY IN TRANSPORT AND THE ENVIRONMENT

BACKGROUND:

The transport sector accounts for 12-13 per cent of energy consumption in Sri Lanka, and 65 per cent of petroleum products consumed in Sri Lanka. Energy consumption in the transport sector grew at 7 per cent in 1989, and estimates of annual gasoline and auto-diesel growth rates to the year 2000 are 6 and 2.5 per cent respectively. Five per cent overall fuel savings may be possible in the transport sector, but there is no programme to promote fuel conservation in the sector. Achieving the potential savings will entail costs, including the costs of engine tuning, maintenance and fleet down time to the vehicle owner, and the costs of rationalized bus schedules in terms of convenience to bus users. These costs must be estimated, in order to determine what type of incentives and regulations would be required to reduce vehicular emissions.

Vehicular air pollution occurs mainly in Colombo, where approximately 300,000 (60 per cent) of the country's registered vehicles are registered. Motor vehicles account for 65 per cent of oil product consumption, but are estimated to produce disproportionate percentages of petroleum combustion products (75 per cent of particulates, 79 per cent of oxides of nitrogen) in Sri Lanka.

Measures by the Government to address vehicular combustion efficiency and pollution are as follows. A 1987 vehicular pollution control study produced pollution control recommendations, which have not been implemented. CEA/NBRO have initiated a 3-year air quality monitoring programme in Colombo, which could serve as a basis for a Colombo air quality database and development of standards.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Devise a practical programme to assist the improvement of efficiency and more complete combustion in vehicles. Study costs involved in improved maintenance and incentives and regulatory tools that would be needed to ensure that an engine tuning center would be used by vehicle/fleet owners, and determine which owners should be targets of a maintenance improvement programme. Popularize engine tuning and efficient driving practices through publicity and assist in developing several engine tuning centers (or stimulating private enterprise to develop them) with the assistance of large fleet operators. It should be noted that 80 per cent of passenger traffic is borne in Sri Lanka by the commercial bus system.	Immediate	25,000 (For support for testing facilities)	M/T&H, CEA, M/P&E,
2 Continue and expand air pollution monitoring in Colombo and other cities as needed. Establish air quality and emissions standards and enact required supporting legislation.	Medium	25,000	CEA, M/T&H, M/P&E

ENVIRONMENTAL EDUCATION

OVERVIEW

The development of education in Sri Lanka compare favourably with the records of other developing countries. Formal education by a system of schools has its origin in the Pre-British period and the traditional system of education has evolved from the pre-Christian era. The culture and religion has a rich tradition of values favourable to environmental conservation. The literacy rate is high (87 per cent) and school going opportunities are available to all in the school going age.

With the gradual emergence of environmental concerns as dominant issues that affect many facets of human civilization or even the very existence of life on earth, the need to strengthen the environmental aspects of education has been accepted as a prime need, the main objectives of environmental education being the development of attitudes, skills and knowledge favourable for environmental conservation, protection and management.

Formal Educational System

Pre-school education in the country has no systematic structure and is functioning in different parts of the island under varying systems of management.

Even though pre-schools education is considered most important in the development of character, lack of an organized system prevents any meaningful intervention, unless a complete overhaul is implemented.

The present curricula for school education has a significant component of environmental education. However, studies have shown that the teaching of these components need to be strengthened. A recent study by NARESA identified that teacher training and the provision of adequate supplementary reading material and other teaching aids a very important requirements among other needs. Revision of curricula to suit the changing needs has to be considered.

The University system has contributed much to environmental education. A post-graduate level course leading to MSc is conducted in the University of Colombo and a course on environmental engineering is conducted at the Moratuwa University. Staff belonging several disciplines in the Universities have answered the need for professional competence on environmental subjects at different levels. However, the need for professional capabilities in some aspects of environmental management is acute and some Universities are now adjusting to serve these needs. For example, a course on Environmental Impact Assessment, leading to a diploma will be introduced in the Kelaniya University shortly. Manpower needs of the present and projections of needs for the future in this sector have not been systematically estimated.

The research work in the tertiary system of education is one area of knowledge generation in Sri Lanka which can be harnessed to address many issues pertaining to the paucity of data and

knowledge on aspects of environmental conservation and protection. An encouragement for research on environment as requirements of either graduate or post graduate study is appropriate as a means to obtain much needed information. Lack of adequate funds and resources limit the work done by researchers in the universities to those that can be undertaken at present with existing resources.

The Non Formal Education

In respect of the need to enhance the environmental awareness of the public, much has been achieved by the Government and non-governmental institutions but improvements are possible and desirable. The developments in the environmental sector with new issues and possible solutions to the problems emerging with increasing frequency, the need to disseminate this information and debate these issues is evident. To the extent that the public understands and develops a consensus by discussion and debate, environmental issues become resolvable in a democratic social system. The role of the media in promoting such a debate and bringing environmental issues become resolvable in a democratic and bringing environmental issues into the agenda of social priorities has to be recognized as most important. The NGOs have developed as centres which can assist the public in their perception and decision making on environmental issues and their role in Sri Lanka, as in many other countries, has been most valuable in this regard. Further improvement in the system available for NGOs, the public and the media to have access to information is desirable.

The cultural background of the country is favourable to environmental conservation, but a conscious effort has to be made to protect the beneficial aspects of our culture to enhance the environment. Projects on Environmental Conservation should be culture-based and communication on related changes should harness as far as possible the traditional means of communication close to the people.

Four project proposals on environmental education now implemented by CEA, with assistance from NORAD, are designed to contribute towards the solution of the main issues in the sphere of environmental education but they fall far short of covering the entirety of the problem.

The main issues that emerge in the sphere of environmental education are the following:

- (1) Environmental components have to be introduced to the teaching at pre school level in the context of the absence of a formal structure;
- (2) The teaching of environmental components in the school syllabi has to be strengthened;
- (3) Manpower needs in the environmental sector have to be ascertained;
- (4) There is an acute need for data and knowledge on many environmental subjects;

- (5) Absence of a Central institution to collate and disseminate environmental subjects;
and
- (6) The traditional system of knowledge and communication should be studied and
harnessed for the improvement of public participation in environmental
conservation and enhancement.

1	ENVIRONMENTAL COMPONENTS HAVE TO BE INTRODUCED TO THE TEACHING AT PRE SCHOOL LEVEL IN THE CONTEXT OF THE ABSENCE OF A FORMAL STRUCTURE OF PRE-SCHOOL EDUCATION			
BACKGROUND: Pre-school education is handled by many institutions and even individuals without any co-ordinating mechanism. Even a formal registration of such schools to enable the collection of basic data is non-existent. There have been many well meaning efforts to train the teachers of these schools and even though the results have been beneficial, a systematic development of the pre-school education has not been achieved. The system therefore, does not provide a format for a meaningful intervention to introduced environmental components.				
RECOMMENDED ACTION		IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Compile a register of pre-schools and arrange for a comprehensive programme of teacher training. This should include preparation and distribution of a teachers' guide.		Immediate	50,000	CEA, DEA

2	THE TEACHING OF ENVIRONMENTAL COMPONENTS IN THE SCHOOL SYLLABI HAS TO BE STRENGTHENED			
BACKGROUND: A survey has revealed that the present syllabi has environmental components but the teaching of these subjects has to be improved by teacher training and providing teaching aids.				
RECOMMENDED ACTION		IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Prepare a package of teaching aids for schools (primary and secondary) and conduct training programme for Master teachers.		Immediate	250,000	CEA, NIE

3	MANPOWER NEEDS IN THE ENVIRONMENTAL SECTOR HAVE TO BE ASCERTAINED			
BACKGROUND: A multiplicity of institutional arrangements have emerged in response to the demands of the environmental sector and these are gradually increasing in numbers and expanding in respect of the range of activities undertaken. These institutions have to be manned by adequately qualified personnel and the shortage of such personnel is already evident. Future demands may even be more exacting and important. The tertiary system of education where the teaching has to be oriented to produce the needs of the professional sector, may adjust its educational programmes to suit the requirements revealed by a systematic survey of the man power needs in the environmental sector.				
RECOMMENDED ACTION		IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Conduct a survey on manpower needs in the environmental sector.		Immediate	200,000	CEA

4

THERE IS AN ACUTE SHORTAGE OF DATA AND KNOWLEDGE ON MANY ENVIRONMENTAL SUBJECTS

BACKGROUND:

In respect of many facets of environment, policy formulation and planning has been hampered due to absence of reliable data and knowledge. As examples wetlands and mineral resources may be cited. In the University system, many research projects are undertaken as requirements of graduate or post graduate study. A system of research grants would enable the authorities to channel research in the universities to environmental subjects of topical interest.

RECOMMENDED ACTION

IMPLEMENTING PRIORITY

COST (US\$)

IMPLEMENTING AGENCY

1 As system of research grants should be developed to encourage research on environmental topics by the universities.

Immediate

250,000

UGC

2. Strengthen the core-course on Environmental Studies at Regional Universities.

Medium

250,000

UGC

5

ABSENCE OF A CENTRAL INSTITUTION TO COLLATE AND DISSEMINATE ENVIRONMENTAL INFORMATION**BACKGROUND:**

The information relevant to environmental emanate from a multiplicity of sources and the number of such sources are on the increase. Collation of environmental information and providing retrieval facilities including fast disseminating capacity is necessary to encourage meaningful public participation on environmental issues. With the shifting of environmental debate to the political arena enlightened public participation is very essential. Better retrieval facilities for environmental information would, in addition, assist researchers, planners and policy makers in their work.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Establish a Centre for Environmental Studies with facilities for retrieval and audio visual aids.	Medium	1,000,000	M/E&PA

6

THE TRADITIONAL SYSTEM OF KNOWLEDGE AND COMMUNICATION SHOULD BE STUDIED AND HARNESSSED FOR THE IMPROVEMENT OF PUBLIC PARTICIPATION IN ENVIRONMENTAL CONSERVATION AND ENHANCEMENT

BACKGROUND:

Culture is generally regarded as the meaning systems that people create around themselves. Therefore any attempt to improve the quality of the environment and enhance environmental awareness has to take place in the cultural context. No systematic attempt has been made as yet to harness the potential afforded by the culture base for environmental enhancement.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Involve the traditional elite (the native physician) and the clergy by a series of educational programmes in environmental awareness programme.	Immediate	100,000	M/E&PA
2 Use the traditional communication media like puppetry, folk music and theatre to disseminate environmental knowledge.	Immediate	50,000	M/E&PA
3 Enhance the capacity of media personnel in communicating environmental knowledge by seminars and issue of literature.	Medium	50,000	M/E&PA
4 Use the regional radio network to encourage a two-way exchange of information on environment by virtue of their proximity to the grass roots level.	Immediate	25,000	M/E&PA

CULTURE

OVERVIEW

Culture defined as the man-made part of man's environment encompasses the entirety of man's activities. The heritage of man, both material and non material valued as worthy of being protected is taken into consideration for the present purpose.

Apart from the ecological monuments which are under the protection and care of the Department of Archaeology, the non-material forms of the national heritage like traditional arts and crafts drama and music are receiving scant attention due to lack of resources to record, preserve and foster. Apart from the gradual loss of this heritage due to changing social conditions and the lack of facilities to identify and preserve them, major disruptions in settlement patterns like the resettlement programmes under new irrigation schemes and inroads in to settled social systems like development of some tourist resorts can accelerate the process of disruption.

The work now being done by the Ministry of Culture Affairs, the Ministry of Small Industries and Ministries handling religions affairs and such other authorities should develop a co-ordinating mechanism for a coherent policy on culture. It can take the form of a Committee on Conservation of National Heritage. The initial function of the Committee should be an endeavour strengthen the present activities on recording and preserving the folklore, arts and crafts, music, etc.

The main issue in the field of culture is:

- (1) Some aspects of culture which are environment-friendly are facing the threat of drastic change in the facet of modern development.

1	SOME ASPECTS OF CULTURE WHICH ARE ENVIRONMENT-FRIENDLY ARE FACING THE THREAT OF DRASTIC CHANGE IN THE FACET OF MODERN DEVELOPMENT																						
BACKGROUND: Large scale re-location of human settlements have taken place specially in relation to irrigation development schemes of the recent past. This has disrupted cultural systems that has been in continuation over a long period and in such events, recording the aspects of culture pertaining to the community is essential in order to prevent its loss for future generations. Permission granted to the extension and development of tourist industry in certain areas has caused severe forms of cultural degradation.																							
<table><tr><th>RECOMMENDED ACTION</th><th>IMPLEMENTING PRIORITY</th><th>COST (US\$)</th><th>IMPLEMENTING AGENCY</th></tr><tr><td>1 Planning and implementation of a programme for harnessing of cultural resources for environmental conservation.</td><td>Immediate and Medium</td><td>250,000</td><td>M/CA&I, M/E&PA</td></tr><tr><td>2 Keeping adequate records of a culture of community that is being disrupted should be considered imperative and be made a part of activities designed to mitigate environmental impact of development programmes.</td><td>Medium</td><td>50,000</td><td>M/CA&I, PCs</td></tr><tr><td>3 In re-settlement planning, measures adopted to re-settle homogeneous cultural groups with least disruption of their way of life should be encouraged.</td><td></td><td></td><td>M/CA&I, M/E&PA</td></tr><tr><td>4 Development of tourist industry should be subjected to EIA procedure which takes into account the cultural heritage of the country and the locality in which development takes place.</td><td></td><td></td><td>M/CA&I, M/E&PA</td></tr></table>				RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY	1 Planning and implementation of a programme for harnessing of cultural resources for environmental conservation.	Immediate and Medium	250,000	M/CA&I, M/E&PA	2 Keeping adequate records of a culture of community that is being disrupted should be considered imperative and be made a part of activities designed to mitigate environmental impact of development programmes.	Medium	50,000	M/CA&I, PCs	3 In re-settlement planning, measures adopted to re-settle homogeneous cultural groups with least disruption of their way of life should be encouraged.			M/CA&I, M/E&PA	4 Development of tourist industry should be subjected to EIA procedure which takes into account the cultural heritage of the country and the locality in which development takes place.			M/CA&I, M/E&PA
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INSTITUTIONAL CAPACITY

OVERVIEW

Article 27 of the 1978 Constitution assigns the Government broad responsibilities to "protect, preserve and improve the environment for the benefit of the community." The National Environment Act (1980) outlined a comprehensive National environmental policy, and established a Central Environmental Authority (CEA) with environmental policy and regulatory functions. There is extensive legislation covering natural resources, which focuses mainly on resource use and the regulation and disposition of State land. For the most part it predates the National Environment Act. Many agencies share control of the use of natural resources -- about three dozen agencies are involved in management of State land alone. Air, water, land and noise pollution control requirements were established in 1988 and primary implementation is the responsibility of the CEA.

Institutions concerned with environment and natural resources necessarily reflect the national conditions and public policies of the nation. Background analyses in the NEAP that support the recommendations for action have highlighted several significant problems concerning the capacity of Sri Lanka's institutions for policy and programme implementation. The three main institutional issues to be addressed under the NEAP are as follows:

- (1) The lack of clear institutional mandates and responsibilities for environmental policy making at the national level;
- (2) Fragmentation and lack of co-ordination among the multiple agencies with land, water, and other natural resource management responsibilities;
- (3) Requirements to focus and strengthen capacity for pollution control and monitoring.

As identified below, each of these issues requires a combination of institution-building actions, including policy analysis, management reforms, better information, technical assistance, and improved staffing and training.

1

ENVIRONMENTAL POLICY-MAKING

BACKGROUND:

Responsibility for national-level environmental policy making is still evolving, including the essential responsibilities for co-ordinating policy development and monitoring policy implementation. A new Ministry of Environment and Parliamentary Affairs (M/E&PA), and a project Ministry of Environment (MoE) were established in 1990, but the latter is not a statutory body. MoE has only one full-time staff, in addition to the Minister and Secretary, and its operational responsibilities remain to be defined. The Central Environment Authority (CEA) is a statutory body advised by the Environmental Council (comprising representatives of the Government, NGOs, and professionals) and is the only line agency within M/E&PA. The establishment of M/E&PA and MoE has brought into question the nature and scope of CEA's mandate, and the practical demands on its resources given its far-reaching responsibilities to implement pollution control and environmental impact assessment. The National Planning Department (NPD) under the Ministry of Policy Planning and Implementation (M/PP&I) is required to ensure that environmental policy guidance is incorporated into the annual Public Investment Programme (PIP), the Government's rolling five-year capital expenditure plan. Other divisions of M/PP&I have evolved responsibilities for assessing economic development activities. M/PP&I lacks environmental specialists, however, and PIPs have contained little guidance on environmental aspects of planning and management to-date.

Pending the Government's resolution of these institutional roles and responsibilities in environmental policy making, a recent innovative measure to improve environmental co-ordination has been the establishment of a broadly-based, high-level National Environment Steering Committee (NESC), under the co-chairmanship of the Secretaries of M/PP&I and M/E&PA, in 1990. The NESC has established broad parameters for the division of responsibilities among M/E&PA, MOE, and the CEA. The two Ministries will handle policy and co-ordination, while CEA will undertake implementation. The Committee meets monthly and is responsible for review of proposed environmental strategies, including this Environmental Action Plan, and donor co-ordination.

Actions have been initiated to strengthen M/PP&I capabilities to integrate environmental economics into economic development policies. Training and technical assistance programmes are being developed under the five-year Natural Resources and Environmental Policy Project (NAREPP) in co-operation with M/PP&I and M/E&PA.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Provide short-term administrative support for NESC to: (a) review proposed environmental strategies and action plans; (b) identify next steps in implementation of approved strategies and action plans; and (c) ensure co-ordination of donor assistance for environment. Staff time for this purpose can be allocated by M/PP&I and/or M/E&PA.	Immediate	30,000	M/E&PA with M/PP&I
2 Government should provide precise policy guidance and clear organizational mandates to address the following national-level environmental policy making issues:			
a) the responsibilities for environmental policy formulation and co-ordination mandates. To reduce overlaps and maximize effective policy development Government needs to identify clearly the objectives, outputs and reporting responsibilities of the national-level agencies concerned with environmental policy and co-ordination. Additional questions to address are, whether there should be a statutory body with an environmental policy and co-ordination mandate, how such an agency should be linked to sectoral environmental units, and what measures are needed to improve the continuity and technical capabilities of the leadership; and	Immediate	50,000	M/E&PA with M/PP&I

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
b) mechanisms for incorporation of environmental factors into the national planning process. Models for linking environmental policies and programmes with the rest of the PIP and the budget process should be developed and implemented. If the PIP is to integrate environmental and economic factors effectively into operations, Government will need to determine the nature and extent of in-house environmental review capability required and the extent to which it should rely capacities of other agencies, including the integration of CEA's environmental impact assessment procedures into economic analyses and decisions.	Immediate	50,000	M/E&PA with M/PP&I
Once Government has provided the needed guidance regarding the environmental policy making and co-ordination mandate and incorporation of environmental considerations into planning and the project approval process, prepare and implement an organizational training and staffing programme for the designated environmental policy making and co-ordination agencies and M/PP&I.	Medium		M/E&PA with M/PP&I

NATURAL RESOURCE MANAGEMENT

BACKGROUND:

Given Government ownership of 80 percent of the country's land, in one form or another, substantial formal legal authority over the use of land, water and other natural resources is vested in the Government. Two significant institutional issues arise from this fact. First, government management of State land is complex and fragmented. Roughly three dozen agencies manage State land, many others are involved in economic development affecting land use or in regulating private sector land use activities, and still more are concerned with water management, for a total of about sixty important land and water management agencies. None has overall authority over land or water uses, and few, if any, have sufficient budget, staff, or regulatory/incentive tools to manage their resources adequately. Moreover, the Government's plan to devolve land management authority to the Provincial Councils has exacerbated uncertainty regarding local governments' responsibilities for natural resource management. Second, encroachment on State land and disregard of the public values of State lands, waters, and other resources is an ongoing, apparently increasing problem. An important response to this problem is to help individuals, families, and villagers become stakeholders in sound natural resource management by granting full ownership to the property they manage.

Effective environmental policy requires not new agencies but actions to strengthen the capacity of a few existing agencies that can best carry out the specific tasks proposed under the NEAP. Problems identified in the NEAP sections on land and water resources emphasize the need for this response. In the fields of wildlife management and forestry agency responsibilities and needs are clear, and the needs of the DWLC and FD can and must be immediately strengthened. Similarly, strengthening the environmental capabilities of the SGC can and should be done by establishing an environmental cell within it. However, with respect to problems of soil erosion, with its multiple causes and effects, and the many other aspects of land and water facing multiple demands, institutional responses are immensely complex, not least because of the many agencies involved. Yet effective natural resource management requires coherent agency policy guidance and clear definitions and co-ordination of agency implementation responsibilities. What entity establishes these policy guidelines and who ensures that they are implemented are critical questions; the M/LI&MD and M/AD&R, for example, have major responsibilities for soil conservation that remain to be sorted out at the national level. A clear process and approach for addressing these kinds of institutional issues is a high priority of the NEAP.

Effective management of natural resources requires far better, and better managed information on priorities and practical responses to guide the programmes of existing agencies. Important steps in this direction have begun under the LUPPD of the M/LI&MD, but further steps are essential. First, information is necessary to identify and map out where environmental problems are most serious, and to assess the most critical impacts -- environmental, social, and economic. Second, information is needed to identify the best management responses to these impacts -- the kinds of educational measures, incentives, and penalties that will be most effective to achieve desired policy results at the local as well as national level. These information requirements, and the failure to consolidate information gathering and analysis capabilities among multiple agencies, are major barriers to effective institutional responses to land and water management. CEA and other studies indicate that the Soil Conservation Act of 1951 has not been effective, but analysis is lacking on the economic effects of soil erosion, the approaches best calculated to induce farmers to conserve soil, how farmers can organize to address these problems, and what kinds of technical assistance is necessary, and how and by whom it should be provided. Similar questions and analytical needs arise regarding chena cultivation, or water use in irrigation areas. Answers to these questions will help determine what institutions should be involved in the solutions, and what legal, economic, or other policy changes are necessary to achieve desired environmental results.

The recommendations below respond to problems of land and water management by addressing the underlying institutional constraints and opportunities. They do so by addressing needs for: a coherent information gathering, analysis, and dissemination programme; specific actions and procedures to strengthen major land management institutions with clear environmental responsibilities and proven capabilities; institutional capabilities for analyses of schemes for education, incentives, and penalties that can achieve land and water management policies.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Strengthen the capability of LUPPD to provide land use planning data and other assistance to the following agencies for the following purposes: (a) to the FD and DWLC to: i) improve protected area management under the joint conservation management plans to be prepared under the NEAP; ii) facilitate improved coverage of areas needing protection under the protected areas network and exchange of land among DWLC, FD and other agencies; (b) to local governments and farmer organizations to help them carry out local planning to address land degradation and other priority local resource management priorities, such as community forestry or watershed management. Guidance will be needed from the Government, in the context of the devolution of land management authority to the Provincial Councils, as to what Provincial Council agencies, AGAs, local officials (grama niladharis) and/or local farmer organizations should receive this assistance; and	Immediate	50,000 100,000	LUPPD of M/LI&MD LUPPD with FD and DWLC (M/LI&MD) LUPPD with Local Governments

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
(c) to the CEA and selected project approving agencies, to ensure incorporation of appropriate land suitability data into the environmental assessment and pollution permit process.	Medium	50,000	LUPPD with CEA, PAAs
(d) to the M/LI&MD and M/E&PA, with mapping to help them identify the extent and significance of chena cultivation, particularly in areas of existing forest with high agricultural potential.		100,000	LUPPD with M/E&PA and M/LI&MD
To ensure LUPPD's capacity to provide the extensive assistance to line agencies that is proposed under the NEAP, prepare and implement a staffing and training programme for LUPPD. Also assist LUPPD to prepare and implement a land use management training programme at provincial, AGA and local levels.	Immediate		
2 Strengthen the capability of DWLC to:			DWLC (M/LI&MD)
(a) initiate and implement land exchange to improve representation of underrepresented habitats, such as Wet Zone and coastal habitats, in the protected area system;		50,000	
(b) jointly manage protected areas with the FD under the conservation management plans to be prepared under the NEAP;		50,000	
(c) manage and co-ordinate a new wetlands protection initiative in co-operation with CEA, CCD, NARA, and other agencies; and		50,000	
(d) improve preservation of elephants and management of elephant-human conflict.		50,000	

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
To ensure DWLC's capacity to undertake the management activities proposed under the NEAP, prepare and implement a staffing and training programme for DWLC, including joint training exercises with LUPPD and FD. Such staffing and training should supplement measures already proposed under DWLC's Five-Year Investment Plan.		20,000	
3 Strengthen the capability of the FD to:	Immediate		FD (M/LI&MD)
(a) prepare and implement community forestry management programmes in selected areas of intense pressure on protected areas in the Wet Zone, including response to problems of limited land tenure in ways that will increase farmer incentives for environmentally sustainable and profitable long-term use of natural resources;		100,000	
(b) jointly manage protected areas and buffer zones with DWLC, under the conservation management plans to be prepared under the NEAP, in close co-operation with local communities having a stake in the use and conservation of natural resources;		50,000	
(c) utilize LUPPD data to facilitate land exchange and develop an improved forestry management strategy in the Dry Zone; and		30,000	
d) control illegal felling and forest fires, and develop an enforcement capability.		100,000	
Assess whether FD staffing and training measures under the on-going FSDP are adequate to support the actions proposed under the NEAP and, if not, prepare and implement a supplementary staffing and training programme.		30,000	

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
4 Establish an Environmental Cell with a staff of two professionals that would review gem mining permit procedures, identify a package of incentives and penalties for improved environmental protection in gem mining, and oversee environmental monitoring of gem mine operations. Prepare and implement a two-year workplan, including staffing and training, for the SGC Environmental Cell.	Immediate	50,000	SGC
5 Strengthen the co-ordinating and policy capabilities of M/LI&MD in the following priority areas:			M/LI&MD
(a) Conservation management: encourage DWLC-FD co-ordination, in particular concerning land exchanges, leading to improved coverage of protected area networks and joint preparation and implementation of conservation management plans;	Medium	10,000	
(b) Soil conservation: prepare a study of soil conservation incentive mechanisms, including educational approaches, and market, full ownership, and other mechanisms, and recommend agencies or other entities to be responsible for soil conservation measures. Recommendations may include, for example, local governments' responsibilities for enforcement of stream and reservoir reservations; local governments in the hill country piloting models for redirecting chena farmers in environmentally sensitive areas to other areas and/or occupations, and rehabilitation/recycling of marginal tea lands; or private entities incorporating soil conservation into planning, investment, or purchasing programmes.	Immediate	100,000	

NATIONAL ENVIRONMENTAL ACTION PLAN

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
The study should review constraints affecting the former Soil Conservation Department and implementation of existing soil conservation legislation, experience gained under past and current watershed management projects, the possible need to establish a Soil Conservation Agency under the M/LI&MD, and procedures for developing new soil conservation approaches involving public education and involvement; and (c) Water management: commission a study of organizational responsibilities and alternatives to address future environmental, economic, and social impacts of intensified water management requirements.	Medium	100,000	M/LI&MD

3

POLLUTION CONTROL**BACKGROUND:**

Enactment and implementation of pollution control regulations are recent developments in Sri Lanka. Comprehensive industrial pollution licensing and environmental assessment requirements were enacted in 1988. CEA began implementation of industrial licensing in 1990, and environmental assessment procedures, which will also be implemented by the CEA, are currently under preparation for application in 1992. Although CEA has had the statutory mandate to establish and oversee pollution control since 1988, the resources available to the CEA to carry out these functions have not increased proportionately. CEA's budget, staff skills and career incentives are inadequate to its new tasks, and in the recent past frequent turnover of leadership, changes of its location within the Government (Cabinet Ministries, most recently from M/PP&I to M/E&PA and MOE) and an unclear division of labor among CEA, M/E&PA and MOE, have further constrained CEA's efforts to address its broad new pollution control and environmental assessment mandates. Mobilization of existing CEA resources, and their strengthening, will be necessary to ensure immediate integration of environmental protection with economic development.

Other Government agencies have capabilities to monitor and analyze air and water pollution from emissions and effluent or from ambient conditions. Co-ordination of laboratory and staff analysis capabilities will enhance the capabilities of the Government in the areas of highest industrial concentration (greater Colombo) and in other major urban areas where domestic and agricultural pollution problems are most significant. Programmes for pollution monitoring of surface and ground waters require clear identification of agency responsibilities, and co-ordination of monitoring programmes and analyses over time. Delegation and co-ordination of water quality monitoring and analysis activities is an increasingly serious need. There are no agencies in the industrial, urban and energy sectors with more comprehensive pollution control or monitoring capabilities and responsibilities than the CEA.

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
1 Identification of the CEA's priority objectives and provision of adequate resources to achieve those objectives are urgently needed. Consideration should be given to focusing CEA's activities and institutional strengthening exclusively on pollution control and environmental assessment in at least the short run. Continuity of leadership and a clear definition of CEA's reporting requirements are urgently needed. The CEA should:	Immediate	100,000	M/E&PA with CEA
(a) prepare a corporate plan, focusing on development of industrial pollution control, environmental assessment, monitoring capabilities and staffing needs;			M/E&PA with CEA

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
(b) review current industrial licensing arrangements, including alternative regulatory and incentive packages, and prepare and implement an action plan to meet licensing requirements in an agreed timeframe; and			M/E&PA with CEA
(c) develop a plan to expand CEA's regulatory capacity to include the energy, urban and mining sectors in a phased manner, which would be implemented in the medium term.			- do -
(e) develop a plan for establishing and co-ordinating a water quality monitoring and analysis system for critical water bodies and ground water resources throughout the country, including a programme for efficient use of laboratories, analysis capabilities, and information.			- do -
2 Upon adoption of a corporate plan for CEA, prepare and implement a staffing and training programme for CEA, to enable CEA to: a) meet its industrial licensing and environmental impact assessment objectives; and b) provide training to Environmental Cells in the energy, urban and mining sectors and co-ordinate EIA and monitoring activities with those cells. Such training should supplement proposed training under NAREPP.	Medium	200,000	- do -
3 Establish an Environmental Cell in CEB. Prepare and implement a two-year workplan, including staffing and training, for the CEB Environmental Cell. The CEB Environmental Cell should be developed and function in close co-operation with the CEA and M/E&PA.	Immediate	50,000	CEB (M/P&E) with CEA

NATIONAL ENVIRONMENTAL ACTION PLAN

RECOMMENDED ACTION	IMPLEMENTING PRIORITY	COST (US\$)	IMPLEMENTING AGENCY
3 Establish an Environmental Planning Cell in UDA. Prepare a two-year workplan, including staffing and training, to enable the Environmental Planning Cell to develop an environmental planning and O+M capability in UDA. This UDA Environmental Cell should be developed and function in close co-operation with the CEA and M/E&PA.	Immediate	30,000	UDA (M/PP&I) with M/E&PA and CEA

NATIONAL ENVIRONMENTAL STEERING COMMITTEE

Composition of the Committee

Mr R Paskaralingam, Secretary Ministry of Policy Planning and Implementation	-	Chairman
Dr D Nesiah, Secretary Ministry of Environment and Parliamentary Affairs	-	Co-Chairman
Mr R Abeyratne, Secretary Ministry of Public Administration, Provincial Council and Home Affairs	-	Member
Mr A A Wijetunga, Secretary Ministry of Lands, Irrigation and Mahaweli Development	-	Member
Mr A R M Jayawardena, Secretary Ministry of Transport and Highways	-	Member
Mr A S Jayawardena, Secretary Ministry of Industries, Science and Technology	-	Member
Mr W D Ailapperuma, Secretary Ministry of Housing and Construction	-	Member
Mr N V K K Weragoda, Secretary Ministry of Fisheries and Aquatic Resources	-	Member
Mr Dixon Nilaweera, Secretary Ministry of Agricultural Development and Research	-	Member
Mr S A Jinadasa, Addl Secretary to the President and Addl Director, Urban Programme Unit	-	Member
Dr Lloyd Fernando, Addl Secretary/PPI & Director General of National Planning	-	Member
Mr Ackiel Mohamed, Secretary Ministry of Power and Energy	-	Member
Secretary to the Minister of Environment	-	Member
Mr G K Amaratunga, Chairman Central Environmental Authority	-	Member
Dr V Ambalavanar, Director General (PI) Ministry of Policy Planning and Implementation	-	Member
Mr S W P Bulankulame, Chairman Urban Development Authority	-	Member
Mr Lakshman R Watawala, Chairman and DG Greater Colombo Economic Commission	-	Member
Mr Ranjit Fernando, General Manager National Development Bank	-	Member
Mr P Illangovan, National Programme Co-ordinator for MEIP	-	Secretary